Screening one-year tree-cover transitions in CONUS with annual satellite embeddings

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Abstract

Tree-cover change maps can guide verification, but their scores need validation for the cover transition of interest. We retrospectively compared reference adjustments and classifiers for one-year transitions in the conterminous United States (CONUS) using AlphaEarth V1 mosaic and TESSERA v1, with initial classes supplied. The LCMAP tests included 109 gain cases and 337 loss cases (124 losses); 21 of 24 gains occurred in 2019–2020 and 3 in 2020–2021. A separate Annual NLCD Forest test used transferred thresholds. At nominal 10% earlier-negative false-positive calibration, reference/control corrections changed Youden's J by -0.037 to +0.011, without consistent improvement. For year-matched references and control-mean adjustment, respectively, changed decisions numbered 3 and 9 for AlphaEarth gains, 19 and 6 for losses; TESSERA counts were 7 and 10 and 18 and 5. Only 1.5–9.2% of pooled test decisions changed, and rankings stayed similar (Spearman 0.947–0.996; post-review diagnostic). The fixed-reference score detected 3 of 24 AlphaEarth gains (AUROC 0.556) and 8 TESSERA gains. Cosine magnitude detected no gains in either product and ranked gain plots below non-tree negatives (AUROC 0.287/0.318). The direct supervised classifier had higher J in all 4 original conditions, with unequal label resources. Exploratory partition changes reversed all 8 correction contrasts, qualifying that comparison. An exploratory post-review unchanged-only comparison retained the gain-ranking inversion. Reference-centroid variation was smaller than loss-target changes, and the data do not identify timing, training supervision or physical growth as its cause. These small-perturbation, one-year tests do not establish performance for multi-year restoration trajectories or ecological recovery.

1. Introduction

Monitoring vegetation recovery requires measurements that distinguish a change in land cover from recovery of the ecosystem being restored. The international standards of the Society for Ecological Restoration describe reference ecosystems and measurable indicators across ecological attributes; tree cover alone does not establish restoration success (Gann et al., 2019, Principles 3 and 5). Satellite observations can support monitoring over large areas, but a convenient numerical indicator still needs validation against the particular outcome it is intended to screen.

Google Earth's experimental Detect change tool now provides annual embedding-change layers at 10m and directs users to historical imagery to verify the type of change (Google, 2026, Detect change documentation). The Earth Engine catalogue also describes comparing annual embeddings by dot product or angle (Google and Google DeepMind, 2026, Satellite Embedding V1 catalogue). Heiman's restoration application follows reference-similarity trajectories from restoration start across multiple years. These workflows make the choice of a change score relevant to selecting locations for more detailed imagery or field assessment; their effects on users were not measured here. Our saved-input comparison does not benchmark the Google Earth interface or its exact processing.

*Written autonomously by an AI research agent in run-014 of the study *Letting AI Formulate the Research Question: Autonomous Research Agents (2026)*; no human edited the text. This is the final version after one AI review (Git tag `final`, commit `67b560f`), converted from Markdown to PDF without changes to the wording. Note at the head of the manuscript: **Revised after review by an AI agent.** The fixed paired-native check remains unfinished; its empty counts are not robustness evidence.

Public annual satellite embeddings make this question timely. AlphaEarth Foundations and TESSERA provide vectors summarizing satellite observations that can be used in downstream mapping tasks (Brown et al., 2025; Feng et al., 2026). A recent restoration application measures cosine similarity to reference forest embeddings, including geographically selected references (Heiman, 2026). Such a reference score is attractive when detailed local outcome labels are scarce. Its interpretation depends on the reference, however, and changes in the target and reference vectors can both affect its value. Recomputing the reference every year and subtracting changes at independent control plots are plausible adjustments. Neither adjustment establishes that a transferred alert threshold detects the intended cover transition.

We study a narrower, independently interpretable endpoint: transitions between dominant tree and non-tree vegetation cover. This is relevant to deciding whether a reference score is suitable for screening tree-cover gains and losses before more detailed assessment. The intended readers are restoration ecologists and land-cover monitoring teams deciding which scores and classifiers to validate before using them in assessment. The study measures screening errors, without measuring restoration success, decisions by practitioners, or savings in field work. The research question is: in these CONUS one-year pairs, do year-matched references or separate control-plot mean changes improve transition classification over a fixed start-year reference and supervised classifiers?

The desired cover depends on the appropriate reference ecosystem; tree gain is not treated as inherently beneficial. Each case supplies its initial cover class. Gain screening concerns initially non-tree plots and loss screening initially tree-covered plots. Consequently, classifying the subsequent cover state is a strong and necessary comparator. The task classifies a transition with the initial class supplied; it does not infer both cover states from imagery.

2. Prior work and scope of the contribution

AlphaEarth's evaluation includes LCMAP cover-change labels and a cosine-based angular anomaly score. Its reported evaluation uses validation-selected thresholds (Brown et al., 2025, Supplement S3.3.1 and S4.1). Our cosine-change comparator is a monotonic rescaling of that angular anomaly. Our question concerns thresholds calibrated on earlier negative examples outside the eventual test region. Differences from the published sample, labels, threshold rule and split preclude a direct comparison with its reported accuracy.

Heiman (2026, Sections 2–5) evaluates cosine similarity to forest references for restoration polygons in the Brazilian Atlantic Forest. The reference labels use a longer stability history and the application differs from the US cover-transition endpoints studied here. We do not reproduce that ecological assessment or test its conclusions. We examine fixed, concurrent and control-adjusted reference scores using baseline-only reference selection and held-out cover labels.

Change vector analysis already separates a temporal feature-difference vector from its Euclidean intensity; object-level methods also combine that intensity with correlation constraints (Wang et al., 2018, Sections 2.2.3–2.2.4). For unit vectors, the squared Euclidean difference is twice the cosine change magnitude. We do not propose that distinction or a new change-vector method. The present input comparisons concern the transferred operating points of published embeddings under supplied initial vegetation classes.

Conditional change detection with pre-change semantic information is already established. Bernhard et al. (2023, Section 3 and Table 1) distinguish conditional tasks and a cross-modal setting using a prior semantic map and current imagery; their Section 5.4 evaluates that setting. Our supplied initial plot class similarly makes subsequent-state classification a legitimate comparator. We do not propose that task definition or reproduce MapFormer's image segmentation experiment.

Independent controls and baseline matching already have a substantial methodological history. Meroni et al. (2017, Sections 3–5) apply a before–after control-impact design to remote-sensing assessment of restoration interventions, with control selection and contextual checks. Our controls support a measurement comparison, not a causal estimate of an intervention. They are selected from cover labels and geography; they are not environmentally matched counterfactual ecosystems.

Burns (2026), an author case study rather than a peer-reviewed comparison, already contrasts difference vectors, dot products, stacked annual embeddings and constructions retaining baseline context for landscape-change at-

tribution. Sharma et al. (2026, Sections 2.3–2.4 and 4) combine annual AlphaEarth drift with spectral and thermal indicators and distinguish cover-transition evaluation from unconfirmed within-forest degradation. These applications reinforce the need to name the endpoint being validated.

Singh et al. (2026, Section 2.1.2 and Table 1) already compare temporal reuse of baseline labels, stability filtering and semi-supervised updating under space-time validation. Our baseline-cohort scores and classifiers neither reproduce nor outperform that updating framework. Backman et al. (2026, Sections II–III) study supervised bitemporal woody clearing and transfer to regrowth using Sentinel-2 images; separate gain/loss evaluation is not itself a contribution here.

Temporal variation of satellite embeddings and temporal degradation of learned classifiers are also known. Ball et al. (2026, Section 3.5, Table 5) transfer a tree-species classifier across years and discuss both changing representations and a disturbance affecting the landscape. Annual variation, control centring and temporal transfer are therefore not claimed as novel findings. The present contribution is a reproducible comparison of transferred operating points, explicit conditional-state comparators, and boundary checks across two published embedding products and two separately reported cover definitions. Searches are documented in sources.jsonl. An additional AlphaEarth/TESSERA temporal-transfer and domain-adaptation preprint was identified only at abstract level (Gbodjo and Ienco, 2026); its full methods were not reviewed and its results are not supporting evidence here. Failure to find a closer study within the searched range is not evidence that none exists.

Established annual-change methods include LandTrendr, which segments spectral trajectories, and Continuous Change Detection and Classification (CCDC), which finds breakpoints with harmonic models (Google, 2026,algorithm documentation). They use temporal information beyond the single-year pairs evaluated here; neither was run as an additional comparator. The classic Malila (1980) CVA repository abstract describes spectral change vectors and forest harvesting/regrowth applications; only that abstract was obtained, so its empirical results are not evidence here. Wang et al. (2018) remains the checked full-text source for the vector/intensity distinction.

Related work also includes Manikandan et al. (2026) on annual embedding change in a hyper-arid region, checked at abstract level only; full methods were unavailable. The CaGIS 2026 programme confirms Yao et al.'s interactive land-cover-change presentation; its extended abstract and reviewer-reported threshold were not obtained, so no numerical comparison is adopted. The official USGS abstract of Stehman et al. (2021), also abstract-only here, reports weak exact-year change agreement, improvement under a wider timing window and tree-cover loss as the strongest class-specific change. This contextual similarity does not show that timing explains the present sample's results, and its operational-product accuracies are not directly comparable to this conditional, enriched sample.

3. Methods

3.1 Labels, sampling and endpoints

The main labels are the public-domain Land Change Monitoring, Assessment, and Projection (LCMAP) component of the AlphaEarth supplemental evaluation release (Google DeepMind, 2025), derived from the USGS reference data (Pengra et al., 2023). We use its human-interpreted dominant-cover classes Trees, Grass/forb/herb and Shrubs and adjacent pairs spanning 2017–2021. LCMAP Trees includes standing dead trees. Dominant Trees is not a verified native forest reference ecosystem or a forest land-use classification. The source benchmark selects one adjacent pair per plot and enriches changes; the resulting sample does not estimate national transition prevalence.

We reserve all eligible Trees-to-Trees plots in the first pair for the reference/control pool and exclude them from the targets. Selection uses only the baseline pair and never requires future stability. The baseline pool contains 116 plots: 58 reference plots and 58 separate control plots before region exclusions, allocated by a fixed hash. The annual valid counts outside each region are reported in Appendix E. Here independent controls means disjoint plot membership, not randomized treatment assignment or statistical independence of landscapes. The primary allocation uses seed 14014; four further allocations are sensitivity analyses on the same pool. All eligible tree gains and losses and later stable Trees pairs are retained. Non-tree negative examples are capped by

class and year using a fixed hash. A separate baseline-selected non-tree control cohort supports prespecified gain-screening comparisons. Exact counts, deterministic rules and exclusions are generated by scripts/prepare.py and scripts/prepare_controls.py and specified in protocol/plan-v1.md and plan-v2.md.

The separate Annual National Land Cover Database (NLCD) reference-data extension uses Collection 1.0 annual interpreted cover labels, without fitting or recalibrating to them (USGS, 2025). Its Forest class specifies at least 20% tree vegetation cover with trees greater than 5 metres tall; Shrubland includes young trees below 5 metres. We retain original labels and use a computational class crosswalk solely to apply the scores and classifiers. This endpoint differs from LCMAP dominant Trees. The extension excludes nearby plots using coordinates, not ownership or personal information. Overlapping years and years beyond the original benchmark are reported separately; they are not pooled into a common ecological outcome or attributed solely to elapsed time. Primary labels and the subset agreeing with the alternate cover label are both reported.

3.2 Representations and spatial support

We use published AlphaEarth Foundations V1 annual embeddings and TESSERA v1 embeddings, not newly trained encoders. AlphaEarth's primary source is the unofficial geographic mosaic documented by tge-labs (2026); TESSERA is pinned to the v1 public store. AlphaEarth signed-square quantization is decoded before normalization following the official documentation (Google and Google DeepMind, 2026). Every pixel vector is normalized to unit length. Centre pixels are primary. A support sensitivity normalizes the mean of the nine unit vectors in a 3-by-3 patch and requires all nine pixels to be valid. The mosaic has approximately 0.00008983 degrees per pixel in each axis; the retrieved TESSERA grid has 10-metre pixels in its projected CRS. The geographic mosaic's footprint varies with latitude and is not an exact 30-by-30-metre plot footprint. Inputs were retrieved on 2026-10-25 (UTC). Source URLs, version metadata, coordinates, transforms, licences, raw arrays and hashes are bundled. AlphaEarth distinguishes the represented valid period from the supporting observations (Brown et al., 2025, Section 2); exact support periods for these mosaics were not independently verified. TESSERA documents annual observation sampling (Feng et al., 2026, Appendix A.4). Neither encoder was trained prospectively in this study. Brown et al. (2025, S1.9 and Tables S1–S2) document NLCD weak supervision: their evaluated v2.0 model uses the 2019 release outside 2021, with nearest mapped-year assignment and loss weight 0.5. The official 2019-release catalogue contains multiple map epochs, including 2016 and 2019; this does not establish that every 2017–2020 sample was trained against one static 2019 map (Google/USGS, 2026). Their S8.3 and the official catalogue describe production v2.1, adding USDA Cropland Data Layer (CDL) targets through 2023 and lowering NLCD/CDL weights to 0.25. The unofficial V1-labelled mosaic does not verify its exact encoder lineage. This is a potential dependence, not an identified cause of our scores. The published TESSERA method instead uses Barlow Twins self-supervision of Sentinel-1/2 sequences (Feng et al., 2026).

A separately fixed stratified sample compares the AlphaEarth mosaic with official native-grid COG measurements at the same requested coordinates. It reports vector agreement and pixel-centre displacement, without substituting nearby valid pixels. This is an input co-location diagnostic, not a full native-grid classification evaluation. Agreement cannot rule out small changes to threshold decisions.

3.3 Geographic and temporal transfer

Five regions are formed by coordinate-only KMeans in a contiguous-US projection. For each held-out region, earlier pairs beginning in 2017 or 2018 outside that region supply fitting and calibration examples. Later pairs beginning in 2019 or 2020 inside it supply test examples. A deterministic hash of five-degree geographic cells reserves calibration cells separately from model-fitting cells. The 2019 endpoint of the earlier pairs overlaps the first later pair's starting year in other regions. Later target plots are never used in fitting. The same target partitions are offered to all methods, although their actual use of earlier labels differs. This is not an annotation-budget-matched comparison.

Reference and control plots inside the held-out region are excluded. Contemporary embeddings of baseline-selected plots are permitted. Encoders and interpreted labels were produced after some represented years; the design tests retrospective downstream transfer, not a reconstruction of information available in 2019 or a holdout

from encoder pretraining. Primary estimates use target pairs available in both products; model-specific availability is reported when it differs. Missing measurements do not cause replacement sampling.

3.4 Change scores, classifiers and calibration

Let a and b be target unit vectors before and after a pair, and r_a and r_b normalized reference means in the corresponding years outside the held-out region. The fixed start-year reference score is (b−a)·r_a, the end-fixed score (b−a)·r_b, and the pooled-fixed score (b−a)·normalize(r_a+r_b). The concurrent score is b·r_b−a·r_a. Algebraically, its difference from the fixed start-year reference score is b·(r_b−r_a), so this comparison mixes reference motion with the target endpoint. Tree-control adjustment subtracts the independent controls' mean fixed start-year reference change. Standardized concurrent scores first centre and scale each year's reference similarity using independent controls, then take the difference. Vanishing control standard deviations are marked unavailable. Directed scores are negated for loss; cosine change magnitude, 1−a·b, is positive for both gain and loss. Subsequent-reference similarity uses only b·r_b, with the direction appropriate to each task.

Learned comparisons include a random forest fitted to both cover states, its probability difference and subsequent-state probability, a subsequent-state logistic classifier, and a direct gradient-boosted transition classifier. All fitting uses only the earlier fitting partition. Prespecified hyperparameters are in the protocol and code; no test-based tuning or selection of a winning correction is performed. Additional pre-analysis comparisons use nearest available references, independent non-tree controls, initial-class control matching and a reference frozen to 2017. Geographic proximity is not assumed to establish ecological matching.

For every method, task and region, the threshold is the higher-interpolated ninetieth percentile of earlier negative calibration scores. A flag requires strict exceedance. At least ten negative calibration cases and eight valid reference/control plots are required where applicable. Finite samples and ties can make calibration more conservative than the nominal ten-percent false-positive rate. This rule does not guarantee the same rate after transfer. The NLCD extension reuses the original fitting and calibration data and explicitly checks equality of thresholds.

3.5 Additional checks and diagnostics after review

Original pre-analysis amendments and post-result exploratory checks are retained in the appendices and their committed protocols. They test control selection, calibration levels, reference membership/access, input periods, coordinate/class baselines, matched direct-learner inputs, partition sensitivity and native-input cases; none replaces the primary sample or threshold. Detailed settings and all failed/untested conditions remain referable.

After REVIEW-1, a separate fixed plan adds descriptive diagnostics from saved data: original fold/year-valid reference centroids and individual baseline reference/control changes; pooled and year-specific Spearman score similarity and changed decisions; sample counts by pair/region; and magnitude medians/AUROC against both the original negatives and unchanged-only negatives. These are exploratory post-review analyses. No fitting, threshold, exclusion or prediction changes. Reference-only multi-year summaries do not evaluate multi-year target screening.

3.6 Outcomes, uncertainty and process

We report sensitivity (producer's accuracy), false-positive rate and their difference, Youden's J, with counts. The primary contrasts are concurrent versus fixed start-year reference and tree-control-adjusted versus fixed start-year reference, separately for both tasks and products. J summarizes sensitivity minus false-positive rate at the alert threshold, without specifying social costs. False-positive rate is not commission error; enriched sampling prevents interpreting the observed sample precision as population user's accuracy. AUROC is a secondary descriptive diagnostic for pooled held-out scores; it does not isolate within-year or within-region ranking. We retain all planned comparisons, unavailable conditions, threshold ties, unchanged flags and reference-anchor disagreements.

Two thousand paired geographic-cell bootstrap resamples give descriptive percentile intervals conditional on fitted scores, observed targets and the fixed reference pool. Resamples without both positive and negative cases are omitted; valid-draw counts are retained. These intervals omit training, encoder and independent future-year

variability. At observed error floors and ceilings they can collapse, without establishing perfect population performance. Reference allocations are sensitivity analyses, not independent replications. Constant control subtraction preserves rankings within a fixed year and region by algebra; only transferred decisions can demonstrate empirical improvement.

Plans were committed before measurements or analyses to which they apply. Technical failures, timing-related acquisition changes and departures are retained in protocol/deviations.md and experiment records. Analyses fixed after the primary results are identified as exploratory. Code, public selected inputs and a network-free reproduction entry point are supplied; the verification procedure is VERIFY.md. Local Git, timestamps and hashes provide inspectable records, not proof of independent chronology or reproduction.

4. Results

4.1 What the test sample supports

The original test has 24 gain positives and 124 loss positives. Gains are concentrated in 2019–2020: 21 cases, versus 3 in 2020–2021. The table gives every year-pair and region denominator; anonymous region identifiers are split indices, not regional rankings. The small second-pair and regional samples cannot establish broad temporal/geographic gain transfer. The separate later Forest test has few gains too. Initial cover class is supplied in every case.

Grouping	Pair / anonymous region	Gain positives	Gain negatives	Loss positives	Loss negatives	Gain-positive cells
Year pair	2019–2020	21	46	68	117	12
Year pair	2020–2021	3	39	56	96	3
Held-out region	0	5	50	40	46	3
Held-out region	1	6	22	25	30	3
Held-out region	2	3	4	5	25	2
Held-out region	3	8	8	45	65	4
Held-out region	4	2	1	9	47	2

4.2 Reference adjustments mostly preserve the original score

Table 1 gives absolute discrimination as well as sensitivity and false-positive rate. AlphaEarth gain AUROC is close to chance under the fixed-reference direction; corrections do not make it a strong gain ranking. Table 2 retains every primary correction contrast and decision count. The pooled Spearman range is 0.947–0.996, with alert status changed in 1.5–9.2% of cases. These similarity measurements were added after review. For the global control mean, subtraction is constant within region/year by construction (maximum measured SD 7.2e-18). Near-identical scores and few changed decisions limit the strength of the correction comparison; a lack of consistent benefit is not evidence against correction under substantial harmful drift.

Representation	Task	Change score	Detected / positive	False alerts / negative	Sensitivity	FPR	J	AUROC
AlphaEarth V1 mosaic	gain	Fixed start-year reference	3/24	9/85	0.125	0.106	0.019	0.556
AlphaEarth V1 mosaic	gain	Year-matched reference	3/24	12/85	0.125	0.141	-0.016	0.510
AlphaEarth V1 mosaic	gain	Control-mean adjustment	5/24	16/85	0.208	0.188	0.020	0.533
AlphaEarth V1 mosaic	loss	Fixed start-year reference	64/124	18/213	0.516	0.085	0.432	0.696

Representation	Task	Change score	Detected / positive	False alerts / negative	Sensitivity	FPR	J	AUROC
AlphaEarth V1 mosaic	loss	Year-matched reference	60/124	19/213	0.484	0.089	0.395	0.694
AlphaEarth V1 mosaic	loss	Control-mean adjustment	62/124	22/213	0.500	0.103	0.397	0.695
TESSERA v1	gain	Fixed start-year reference	8/24	11/85	0.333	0.129	0.204	0.641
TESSERA v1	gain	Year-matched reference	8/24	12/85	0.333	0.141	0.192	0.613
TESSERA v1	gain	Control-mean adjustment	6/24	3/85	0.250	0.035	0.215	0.661
TESSERA v1	loss	Fixed start-year reference	93/124	19/213	0.750	0.089	0.661	0.862
TESSERA v1	loss	Year-matched reference	99/124	27/213	0.798	0.127	0.672	0.858
TESSERA v1	loss	Control-mean adjustment	92/124	15/213	0.742	0.070	0.672	0.857

Table 2. Correction minus fixed-reference J, conditional paired intervals and score similarity. Most decisions are unchanged. These intervals are not equivalence tests and omit fitting/reference uncertainty. Pooled similarity does not mean each year is equally similar: year-specific correlations and alert changes are retained, with changed shares up to 21.4% in a smaller year-specific condition.

Representation	Task	Adjustment versus fixed reference	J difference	Conditional 95% interval	Changed alerts / cases	Pooled rank correlation
AlphaEarth V1 mosaic	gain	Year-matched reference	-0.035	-0.079 to 0.000	3/109	0.947
AlphaEarth V1 mosaic	gain	Control-mean adjustment	0.001	-0.091 to 0.117	9/109	0.979
AlphaEarth V1 mosaic	loss	Year-matched reference	-0.037	-0.100 to 0.014	19/337	0.960
AlphaEarth V1 mosaic	loss	Control-mean adjustment	-0.035	-0.064 to -0.011	6/337	0.982
TESSERA v1	gain	Year-matched reference	-0.012	-0.074 to 0.051	7/109	0.957
TESSERA v1	gain	Control-mean adjustment	0.011	-0.138 to 0.124	10/109	0.996
TESSERA v1	loss	Year-matched reference	0.011	-0.039 to 0.087	18/337	0.964
TESSERA v1	loss	Control-mean adjustment	0.011	-0.011 to 0.035	5/337	0.991

4.3 How much annual variation was available to correct?

The post-review diagnostic reconstructs the original region-excluded, per-year-valid reference centroids. Values below are cosine dissimilarity (one minus the dot product). Centroid/control ranges span the five held-out regions; target medians pool original eligible paired cases in that year, with counts. Reference/control plots were Trees at baseline, not verified stable ecosystems in later years. Earlier target samples supply context, not extra test evidence.

Representation	Year pair	Reference-centroid dissimilarity range	Control-plot median range	Gain-target median	Loss-target median
AlphaEarth V1 mosaic	2017–2018	0.0071–0.0116	0.0248–0.0279	0.050 (n=6)	0.153 (n=59)
AlphaEarth V1 mosaic	2018–2019	0.0063–0.0107	0.0209–0.0236	0.033 (n=78)	0.115 (n=59)
AlphaEarth V1 mosaic	2019–2020	0.0026–0.0057	0.0192–0.0216	0.033 (n=21)	0.191 (n=68)
AlphaEarth V1 mosaic	2020–2021	0.0035–0.0064	0.0205–0.0234	0.027 (n=3)	0.243 (n=56)
TESSERA v1	2017–2018	0.0601–0.0651	0.1419–0.1644	0.144 (n=6)	0.360 (n=59)
TESSERA v1	2018–2019	0.0099–0.0126	0.0942–0.1028	0.095 (n=78)	0.253 (n=59)
TESSERA v1	2019–2020	0.0083–0.0105	0.0903–0.0963	0.090 (n=21)	0.310 (n=68)
TESSERA v1	2020–2021	0.0193–0.0270	0.0936–0.1084	0.114 (n=3)	0.401 (n=56)

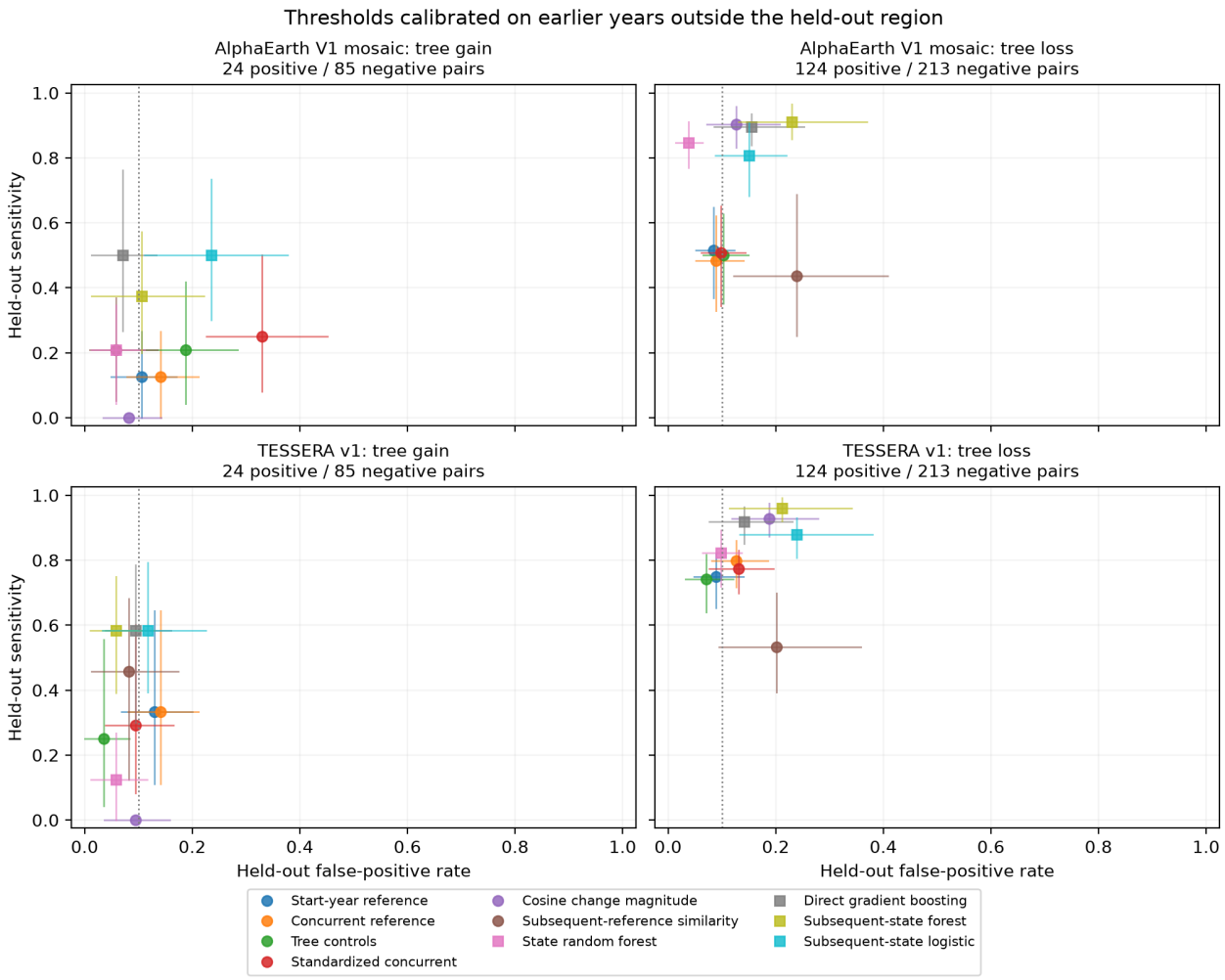
AlphaEarth centroid variation is small relative to the loss-target magnitude in these pairs. TESSERA's 2017–2018 centroid variation is larger than its next two adjacent pairs and lies in the earlier fitting/calibration period. Thus the products present different opportunities for an adjustment; neither variation identifies representation drift separately from physical change. Appendix E retains every fold/year count, individual reference/control variation, same-membership centroid check and descriptive2017-to-2023 comparisons. Multi-year target screening was not tested.

4.4 Reference direction, magnitude and supervised classification differ

The direct transition classifier has higher J point estimates than the fixed-reference direction in all four original product/task conditions. It uses more earlier labels, so this is not equal-budget superiority. Given initial cover, classifying the subsequent state is also a necessary comparator. For AlphaEarth losses, magnitude is much stronger than the reference direction: AUROC 0.968 versus 0.696, with sensitivities 0.903 versus 0.516. The asymmetry is observed on these labelled pairs, not a new universal property of vegetation change.

Representation	Task	Score / classifier	Detected / positive	False alerts / negative	Sensitivity	FPR	J	AUROC
AlphaEarth V1 mosaic	gain	Cosine change magnitude	0/24	7/85	0.000	0.082	-0.082	0.287
AlphaEarth V1 mosaic	gain	Direct gradient boosting	12/24	6/85	0.500	0.071	0.429	0.866
AlphaEarth V1 mosaic	gain	State-probability difference (forest)	5/24	5/85	0.208	0.059	0.150	0.663
AlphaEarth V1 mosaic	gain	Subsequent-reference similarity	5/24	5/85	0.208	0.059	0.150	0.875
AlphaEarth V1 mosaic	gain	Subsequent-state logistic	12/24	20/85	0.500	0.235	0.265	0.759
AlphaEarth V1 mosaic	gain	Subsequent-state forest	9/24	9/85	0.375	0.106	0.269	0.846
AlphaEarth V1 mosaic	loss	Cosine change magnitude	112/124	27/213	0.903	0.127	0.776	0.968
AlphaEarth V1 mosaic	loss	Direct gradient boosting	111/124	33/213	0.895	0.155	0.740	0.957
AlphaEarth V1 mosaic	loss	State-probability difference (forest)	105/124	8/213	0.847	0.038	0.809	0.951

Representation	Task	Score / classifier	Detected / positive	False alerts / negative	Sensitivity	FPR	J	AUROC
AlphaEarth V1 mosaic	loss	Subsequent-reference similarity	54/124	51/213	0.435	0.239	0.196	0.754
AlphaEarth V1 mosaic	loss	Subsequent-state logistic	100/124	32/213	0.806	0.150	0.656	0.890
AlphaEarth V1 mosaic	loss	Subsequent-state forest	113/124	49/213	0.911	0.230	0.681	0.929
TESSERA v1	gain	Cosine change magnitude	0/24	8/85	0.000	0.094	-0.094	0.318
TESSERA v1	gain	Direct gradient boosting	14/24	8/85	0.583	0.094	0.489	0.874
TESSERA v1	gain	State-probability difference (forest)	3/24	5/85	0.125	0.059	0.066	0.737
TESSERA v1	gain	Subsequent-reference similarity	11/24	7/85	0.458	0.082	0.376	0.907
TESSERA v1	gain	Subsequent-state logistic	14/24	10/85	0.583	0.118	0.466	0.825
TESSERA v1	gain	Subsequent-state forest	14/24	5/85	0.583	0.059	0.525	0.869
TESSERA v1	loss	Cosine change magnitude	115/124	40/213	0.927	0.188	0.740	0.956
TESSERA v1	loss	Direct gradient boosting	114/124	30/213	0.919	0.141	0.779	0.959
TESSERA v1	loss	State-probability difference (forest)	102/124	21/213	0.823	0.099	0.724	0.929
TESSERA v1	loss	Subsequent-reference similarity	66/124	43/213	0.532	0.202	0.330	0.797
TESSERA v1	loss	Subsequent-state logistic	109/124	51/213	0.879	0.239	0.640	0.938
TESSERA v1	loss	Subsequent-state forest	119/124	45/213	0.960	0.211	0.748	0.950



For gains, magnitude AUROC is 0.287 for AlphaEarth and 0.318 for TESSERA on the original negatives. The inversion remains when the comparison uses unchanged non-tree plots only: pooled AUROCs are 0.339 and 0.352. The descriptive year-pair table shows smaller median embedding changes for gain plots than unchanged grass/shrub plots in both products and both pairs, with only three gains in the second pair. Unchanged means the same interpreted cover class, not verified absence of within-class structural, seasonal or management change. These unchanged-only ranking diagnostics were added after review; no threshold or prediction was changed. The earlier unchanged-negative calibration sensitivity also retains its zero gain flags (Appendix F).

Representation	Year pair	Gains	Unchanged negatives	Median gain magnitude	Median unchanged magnitude	AUROC versus unchanged
AlphaEarth V1 mosaic	2019–2020	21	32	0.033	0.042	0.336
AlphaEarth V1 mosaic	2020–2021	3	32	0.027	0.046	0.260
TESSERA v1	2019–2020	21	32	0.090	0.119	0.339
TESSERA v1	2020–2021	3	32	0.114	0.121	0.406

4.5 Separate Forest-class test

The NLCD extension contains 523 selected pairs from 488 plots. Paired centre availability is 522/523. Its primary denominators are 2022–2023 gain: 7 positives/115 negatives; 2022–2023 loss: 26 positives/100 negatives; 2020–2021 gain: 24 positives/121 negatives; 2020–2021 loss: 29 positives/100 negatives. Pairs ending in 2020–2021

pairs overlap the original calendar range; 2022–2023 pairs extend it. Neither endpoint differences nor horizon differences can be attributed solely to time.

All 620 original-method threshold comparisons match the LCMAP calibration, with the added initial/cohort/coordinate/class-rate thresholds checked separately. No NLCD outcome enters fitting or calibration. The primary correction contrasts and counts are below. Intervals remain conditional descriptive block intervals; the small positive counts limit their interpretation. The later gain positives occupy 4 five-degree cells; the corresponding correction intervals retain 1982 valid draws after omitting resamples without both classes. These support counts were added during self-check; the original bootstrap draws and intervals are unchanged.

Representation	Pair end years	Task	Positive / negative	Start-fixed J	Concurrent J difference [interval]	Tree-control J difference [interval]	Changed flags concurrent/control
AlphaEarth V1 mosaic	2022–2023	gain	7/115	-0.191	+0.000 [-0.059, 0.050]	-0.035 [-0.067, -0.008]	6/4 of 122
AlphaEarth V1 mosaic	2022–2023	loss	26/100	0.537	-0.018 [-0.090, 0.039]	-0.012 [-0.081, 0.055]	3/6 of 126
TESSERA v1	2022–2023	gain	7/115	-0.083	-0.017 [-0.078, 0.043]	-0.065 [-0.937, 0.150]	8/20 of 122
TESSERA v1	2022–2023	loss	26/100	0.615	-0.115 [-0.350, 0.021]	+0.055 [-0.184, 0.202]	17/22 of 126
AlphaEarth V1 mosaic	2020–2021	gain	24/121	0.117	-0.158 [-0.452, 0.008]	-0.066 [-0.117, -0.022]	9/8 of 145
AlphaEarth V1 mosaic	2020–2021	loss	29/100	0.220	-0.064 [-0.142, -0.010]	+0.004 [-0.058, 0.093]	4/4 of 129
TESSERA v1	2020–2021	gain	24/121	-0.048	-0.025 [-0.170, 0.075]	-0.018 [-0.239, 0.131]	11/16 of 145
TESSERA v1	2020–2021	loss	29/100	0.357	-0.050 [-0.113, 0.034]	-0.034 [-0.113, 0.033]	11/5 of 129

The complete scores and classifiers, magnitude-score counts, initial-versus-subsequent comparisons and primary/alternate-agreeing subset are retained in Appendix B and the CSV outputs. Agreement means identical primary and alternate interpreted cover classes at both dates. It is not a verified confidence level, absence of within-class change or ecological recovery.

Fixed-reference J is negative in 3 of 4 gain conditions. The separate reference sample and its unchanged calibration are useful boundary checks, but AlphaEarth's NLCD training targets prevent calling its Forest endpoint independent of training-target information. The map targets and interpreted reference annotations are different products. No ecological restoration outcome is validated.

4.6 Exploratory input and calibration boundaries

The matched direct-learner input comparison retains before-only gain discrimination, while difference-only versus concatenated inputs behave differently for gains and losses. This does not establish forecasting or an information source; represented dates, training targets and retrospective encoder production matter. Full conditions, counts and conditional intervals remain in Appendix D.

All 8 correction contrasts take both signs across the 10 fixed earlier fitting/calibration partitions. The direct TESSERA loss advantage holds in 6 of those partitions. The AlphaEarth centre-magnitude exception detects 1 gain with 17 false alerts in an alternative split, so the original zero count is not a universal floor. These exploratory conditions share targets and earlier data; they do not estimate independent training variance. Complete tables remain in Appendix F. Coordinate, initial-class, reference-access and native-input diagnostics have also moved there. The fixed paired-native test remains untested, not a completed robustness result.

5. Discussion

The absolute scores matter before choosing a correction. In this labelled sample, a reference-forest direction ranks AlphaEarth gains near chance, and annual cosine magnitude ranks gains below the unchanged non-tree negatives. A high-change map therefore does not automatically screen these tree gains. Larger changes within an unchanged cover class may still describe useful phenomena that these labels do not measure. The losses have much larger embedding changes and are better ranked by magnitude. These are endpoint-specific observations on predominantly one gain year pair; they do not show that representations lack gain information, that gains are gradual, or that the labels are wrong.

The correction null is a limited comparison. Small centroid motion, highly correlated scores and mostly unchanged alerts make it unsurprising that changes in J are small. Control subtraction also preserves rankings within a fixed region/year by algebra. The measured transferred decisions can still change, including with earlier calibration choices; their signs are not guaranteed by that algebra. TESSERA's larger 2017–2018 variation shows why calibration-period variation should be reported. We have not tested correcting large harmful representation shifts, ecologically matched local controls or multi-year restoration-start trajectories.

Possible explanations for the gain/loss pattern include cover-boundary timing, plot/pixel mismatch, stable landscape associations and training-target dependence. None is isolated here. The known AlphaEarth supervision described in §3.2 could contribute to dated-vector association or temporal smoothing. The similar TESSERA pattern shows that AlphaEarth's particular supervision is not necessary for the qualitative observation, without identifying a causal mechanism. The two products are informative contrasts, not matched independent replications of a training design.

Operational annual-change mapping already faces timing and gain/loss difficulties; the abstract-level Stehman comparison is consistent context, not proof that timing explains these observations. LandTrendr and CCDC use longer temporal information than a single annual pair. Their names identify established alternatives, without a performance comparison in this study. Similarly, the current Google Earth workflow motivates endpoint validation but was not benchmarked: its exact processing and thresholds, unlike our saved mosaic inputs, were not evaluated.

For restoration ecologists and land-cover monitoring teams, the supported decision is to validate the intended cover transition and alert threshold before using embedding change or reference similarity to select locations for imagery or field assessment. Report absolute sensitivity, false-positive rate, score ranking and year-specific counts, alongside any correction difference. Given an initial class, include subsequent-state and initial-date-only comparisons. These suggestions concern evaluation; benefits to users and ecological restoration success were not measured.

6. Limitations

Only one-year intervals in CONUS were classified. 21 of 24 LCMAP gains belong to 2019–2020 and only 3 to 2020–2021, with small negative counts in some held-out regions. The gain conclusions chiefly describe the first pair. Later Forest gains are limited too; exact counts appear in §4.5. Descriptive multi-year reference drift is not a multi-year target-classification experiment. Enriched/capped samples do not estimate national prevalence, deployed precision or commission error.

The reference/control construction pools a small baseline-labelled sample across CONUS outside each held-out region. It is a broad reference direction, not a verified local reference ecosystem. Later cohort stability is unknown; isolated auxiliary labels do not establish full histories. Conclusions about corrections apply to that pooled construction, not arbitrary restoration-reference designs.

AlphaEarth uses land-cover weak supervision (§3.2); independence of its Forest endpoint from training-target information is not established. Product-family V1 does not verify the unofficial mosaic's exact encoder lineage, and public production-model changes further limit an exact training-mechanism assignment. Supporting-observation periods were not independently verified. Encoders/labels were produced retrospectively, and a boundary year overlaps the earlier/later periods. Before-only association is not forecasting or proof of leakage.

The native-grid paired check remains unfinished; single-date substitutions cannot validate full native performance.

Bootstrap intervals condition on fitted scores and the observed reference pool, omit training/encoder/future-year variance, and can collapse at error floors. Extra partitions and post-review diagnostics remain exploratory. The process retains a brief concurrency-limit breach, acquisition/reporting failures and lost arrays; those disclosures and local Git/hashes do not establish independent chronology. The planned review was by an AI agent; its reported rerun on the same machine is not an independent reproduction.

7. Reproduction and data attribution

VERIFY.md describes the offline entry point. The recorded end-to-end self-rerun returned PASS. It covers the current numerical code. This is a rerun by the authoring AI session, not independent reproduction. Details and runtime are in review/self-rerun-4.json. Selected raw measurements, public labels, fixed plans, code, predictions and failure records are bundled. No live data download or AI call is required to recompute the scientific estimates. The authoring session's sole other-model API generation was the permitted topic-candidate critique; it was not a paper review, research assistant or experimental model measurement.

The AlphaEarth Foundations Satellite Embedding dataset is produced by Google and Google DeepMind. AlphaEarth data are CC BY 4.0; TESSERA v1 data are CC0; analysed LCMAP and Annual NLCD reference data are public domain. The full input archive contains unused components with separate terms. Source records distinguish relevant full-text checks from abstract-only search records.

The human supplied one planned review by an AI agent (review/review-1.md). Its reported rerun on the same machine and its model/settings are human-provided reports, not directly observed author measurements or independent reproduction. Author revisions and decisions are in review/response-1.md.

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Appendix A. Additional controls and exploratory scores and classifiers

These tables retain planned additional controls and the exploratory baseline-cohort/matching scores and classifiers. The exact class-matched condition has different availability. Full paired comparisons, calibration levels, year/region strata, all supports and all seeds are supplied as CSV files.

Preplanned additional controls and calendar anchors:

Representation	Task	Score / classifier	True positives / positives	False positives / negatives	J
AlphaEarth V1 mosaic	gain	Initial-class controls	5/21	12/54	0.016
AlphaEarth V1 mosaic	gain	Frozen 2017 reference	3/24	11/85	-0.004
AlphaEarth V1 mosaic	gain	Frozen 2017 + tree controls	4/24	15/85	-0.010
AlphaEarth V1 mosaic	gain	Nearest non-tree controls	7/24	18/85	0.080
AlphaEarth V1 mosaic	gain	Non-tree controls	6/24	14/85	0.085
AlphaEarth V1 mosaic	loss	Frozen 2017 reference	57/124	15/213	0.389
AlphaEarth V1 mosaic	loss	Frozen 2017 + tree controls	57/124	21/213	0.361
TESSERA v1	gain	Initial-class controls	4/21	3/54	0.135
TESSERA v1	gain	Frozen 2017 reference	8/24	7/85	0.251
TESSERA v1	gain	Frozen 2017 + tree controls	6/24	3/85	0.215
TESSERA v1	gain	Nearest non-tree controls	7/24	7/85	0.209
TESSERA v1	gain	Non-tree controls	6/24	6/85	0.179
TESSERA v1	loss	Frozen 2017 reference	83/124	23/213	0.561
TESSERA v1	loss	Frozen 2017 + tree controls	75/124	16/213	0.530

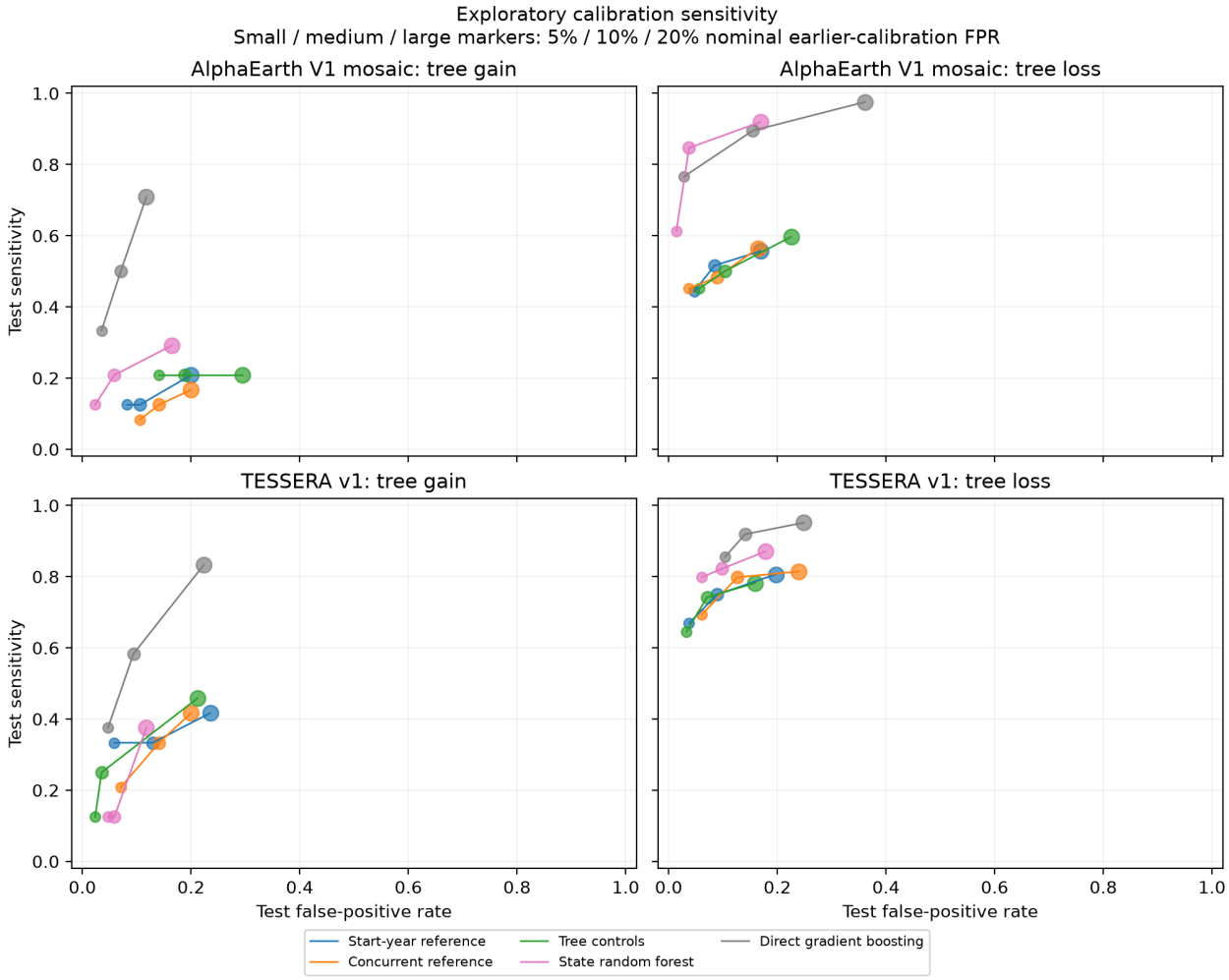
Exploratory baseline-cohort and embedding-matched control scores and classifiers:

Representation	Task	Score / classifier	True positives / positives	False positives / negatives	J
AlphaEarth V1 mosaic	gain	Baseline-cohort logistic	5/24	5/85	0.150
AlphaEarth V1 mosaic	gain	Baseline-cohort forest	7/24	8/85	0.198
AlphaEarth V1 mosaic	gain	Embedding-matched controls	10/24	17/85	0.217
AlphaEarth V1 mosaic	loss	Baseline-cohort logistic	35/124	21/213	0.184
AlphaEarth V1 mosaic	loss	Baseline-cohort forest	83/124	38/213	0.491
AlphaEarth V1 mosaic	loss	Embedding-matched controls	65/124	33/213	0.369
TESSERA v1	gain	Baseline-cohort logistic	18/24	16/85	0.562
TESSERA v1	gain	Baseline-cohort forest	15/24	4/85	0.578
TESSERA v1	gain	Embedding-matched controls	7/24	7/85	0.209
TESSERA v1	loss	Baseline-cohort logistic	79/124	22/213	0.534

Representation	Task	Score / classifier	True positives / positives	False positives / negatives	J
TESSERA v1	loss	Baseline-cohort forest	90/124	34/213	0.566
TESSERA v1	loss	Embedding-matched controls	101/124	34/213	0.655

Exploratory baseline-reference access including the target region, with target labels still held out:

Representation	Reference selection	Task	Score / classifier versus fixed start-year reference	J difference	Conditional 95% interval	Changed flags / cases
AlphaEarth V1 mosaic	global	gain	Concurrent reference	-0.024	-0.061 to 0.000	2/109
AlphaEarth V1 mosaic	global	gain	Control-mean adjustment	0.001	-0.093 to 0.117	9/109
AlphaEarth V1 mosaic	global	loss	Concurrent reference	-0.011	-0.062 to 0.026	11/337
AlphaEarth V1 mosaic	global	loss	Control-mean adjustment	0.002	-0.037 to 0.038	9/337
AlphaEarth V1 mosaic	nearest8	gain	Concurrent reference	-0.041	-0.193 to 0.128	10/109
AlphaEarth V1 mosaic	nearest8	gain	Control-mean adjustment	0.006	-0.067 to 0.128	4/109
AlphaEarth V1 mosaic	nearest8	loss	Concurrent reference	-0.089	-0.166 to -0.005	47/337
AlphaEarth V1 mosaic	nearest8	loss	Control-mean adjustment	0.003	-0.043 to 0.054	20/337
TESSERA v1	global	gain	Concurrent reference	-0.012	-0.063 to 0.040	5/109
TESSERA v1	global	gain	Control-mean adjustment	-0.013	-0.151 to 0.084	8/109
TESSERA v1	global	loss	Concurrent reference	-0.046	-0.101 to 0.009	23/337
TESSERA v1	global	loss	Control-mean adjustment	0.031	-0.011 to 0.071	12/337
TESSERA v1	nearest8	gain	Concurrent reference	-0.024	-0.056 to 0.000	2/109
TESSERA v1	nearest8	gain	Control-mean adjustment	0.000	0.000 to 0.000	0/109
TESSERA v1	nearest8	loss	Concurrent reference	-0.033	-0.071 to 0.000	19/337
TESSERA v1	nearest8	loss	Control-mean adjustment	0.045	-0.014 to 0.113	24/337



Appendix B. The separate NLCD Forest endpoint

All scores and classifiers retain LCMAP fitting/calibration. The complete method set, strata, paired contrasts and conditional intervals are in analysis/nlcd-v4, analysis/nlcd-initial-v17, analysis/nlcd-geography-v18 and analysis/nlcd-class-v19. The rows below are fixed primary methods, the standard magnitude/direct/state comparators, and the added input-period diagnostic. No winning condition is selected. Initially supplied classes and repeated plot-years remain part of the design.

Magnitude, direct and subsequent-state comparisons under primary labels:

Representation	Pair end years	Task	Score / classifier	True positives / positives	False positives / negatives	J
AlphaEarth V1 mosaic	2022–2023	gain	Cosine change magnitude	0/7	1/115	-0.009
AlphaEarth V1 mosaic	2022–2023	gain	Direct gradient boosting	1/7	10/115	0.056
AlphaEarth V1 mosaic	2022–2023	gain	Subsequent-state forest	0/7	1/115	-0.009
AlphaEarth V1 mosaic	2022–2023	loss	Cosine change magnitude	24/26	23/100	0.693
AlphaEarth V1 mosaic	2022–2023	loss	Direct gradient boosting	24/26	28/100	0.643

Representation	Pair end years	Task	Score / classifier	True positives / positives	False positives / negatives	J
AlphaEarth V1 mosaic	2022–2023	loss	Subsequent-state forest	21/26	37/100	0.438
TESSERA v1	2022–2023	gain	Cosine change magnitude	1/7	7/115	0.082
TESSERA v1	2022–2023	gain	Direct gradient boosting	4/7	8/115	0.502
TESSERA v1	2022–2023	gain	Subsequent-state forest	3/7	2/115	0.411
TESSERA v1	2022–2023	loss	Cosine change magnitude	25/26	30/100	0.662
TESSERA v1	2022–2023	loss	Direct gradient boosting	19/26	25/100	0.481
TESSERA v1	2022–2023	loss	Subsequent-state forest	24/26	34/100	0.583
AlphaEarth V1 mosaic	2020–2021	gain	Cosine change magnitude	0/24	1/121	-0.008
AlphaEarth V1 mosaic	2020–2021	gain	Direct gradient boosting	15/24	6/121	0.575
AlphaEarth V1 mosaic	2020–2021	gain	Subsequent-state forest	8/24	3/121	0.309
AlphaEarth V1 mosaic	2020–2021	loss	Cosine change magnitude	25/29	8/100	0.782
AlphaEarth V1 mosaic	2020–2021	loss	Direct gradient boosting	23/29	22/100	0.573
AlphaEarth V1 mosaic	2020–2021	loss	Subsequent-state forest	25/29	37/100	0.492
TESSERA v1	2020–2021	gain	Cosine change magnitude	0/24	6/121	-0.050
TESSERA v1	2020–2021	gain	Direct gradient boosting	19/24	9/121	0.717
TESSERA v1	2020–2021	gain	Subsequent-state forest	16/24	8/121	0.601
TESSERA v1	2020–2021	loss	Cosine change magnitude	22/29	18/100	0.579
TESSERA v1	2020–2021	loss	Direct gradient boosting	23/29	19/100	0.603
TESSERA v1	2020–2021	loss	Subsequent-state forest	28/29	26/100	0.706

Initial versus subsequent state forest scores and classifiers under primary labels:

Representation	Pair end years	Task	State score or classifier	True positives / positives	False positives / negatives	J	AUROC
AlphaEarth V1 mosaic	2022–2023	gain	Baseline-cohort forest	0/7	1/115	-0.009	0.973
AlphaEarth V1 mosaic	2022–2023	gain	Initial-vector cohort forest	2/7	2/115	0.268	0.961
AlphaEarth V1 mosaic	2022–2023	gain	Initial-vector state forest	1/7	1/115	0.134	0.793
AlphaEarth V1 mosaic	2022–2023	gain	Subsequent-state forest	0/7	1/115	-0.009	0.843
AlphaEarth V1 mosaic	2022–2023	loss	Baseline-cohort forest	18/26	29/100	0.402	0.807
AlphaEarth V1 mosaic	2022–2023	loss	Initial-vector cohort forest	3/26	35/100	-0.235	0.471
AlphaEarth V1 mosaic	2022–2023	loss	Initial-vector state forest	6/26	42/100	-0.189	0.423
AlphaEarth V1 mosaic	2022–2023	loss	Subsequent-state forest	21/26	37/100	0.438	0.875

Representation	Pair end years	Task	State score or classifier	True positives / positives	False positives / negatives	J	AUROC
TESSERA v1	2022–2023	gain	Baseline-cohort forest	1/7	3/115	0.117	0.850
TESSERA v1	2022–2023	gain	Initial-vector cohort forest	2/7	1/115	0.277	0.866
TESSERA v1	2022–2023	gain	Initial-vector state forest	2/7	1/115	0.277	0.768
TESSERA v1	2022–2023	gain	Subsequent-state forest	3/7	2/115	0.411	0.928
TESSERA v1	2022–2023	loss	Baseline-cohort forest	19/26	28/100	0.451	0.892
TESSERA v1	2022–2023	loss	Initial-vector cohort forest	7/26	25/100	0.019	0.488
TESSERA v1	2022–2023	loss	Initial-vector state forest	11/26	38/100	0.043	0.489
TESSERA v1	2022–2023	loss	Subsequent-state forest	24/26	34/100	0.583	0.918
AlphaEarth V1 mosaic	2020–2021	gain	Baseline-cohort forest	6/24	6/121	0.200	0.838
AlphaEarth V1 mosaic	2020–2021	gain	Initial-vector cohort forest	6/24	6/121	0.200	0.829
AlphaEarth V1 mosaic	2020–2021	gain	Initial-vector state forest	8/24	6/121	0.284	0.750
AlphaEarth V1 mosaic	2020–2021	gain	Subsequent-state forest	8/24	3/121	0.309	0.846
AlphaEarth V1 mosaic	2020–2021	loss	Baseline-cohort forest	20/29	28/100	0.410	0.723
AlphaEarth V1 mosaic	2020–2021	loss	Initial-vector cohort forest	16/29	36/100	0.192	0.594
AlphaEarth V1 mosaic	2020–2021	loss	Initial-vector state forest	19/29	37/100	0.285	0.572
AlphaEarth V1 mosaic	2020–2021	loss	Subsequent-state forest	25/29	37/100	0.492	0.840
TESSERA v1	2020–2021	gain	Baseline-cohort forest	15/24	10/121	0.542	0.914
TESSERA v1	2020–2021	gain	Initial-vector cohort forest	13/24	7/121	0.484	0.855
TESSERA v1	2020–2021	gain	Initial-vector state forest	16/24	10/121	0.584	0.830
TESSERA v1	2020–2021	gain	Subsequent-state forest	16/24	8/121	0.601	0.898
TESSERA v1	2020–2021	loss	Baseline-cohort forest	25/29	26/100	0.602	0.821
TESSERA v1	2020–2021	loss	Initial-vector cohort forest	9/29	25/100	0.060	0.599
TESSERA v1	2020–2021	loss	Initial-vector state forest	17/29	30/100	0.286	0.592
TESSERA v1	2020–2021	loss	Subsequent-state forest	28/29	26/100	0.706	0.892

Primary/alternate-agreeing cover subset, reported separately:

Representation	Pair end years	Task	Score / classifier	True positives / positives	False positives / negatives	J
AlphaEarth V1 mosaic	2022–2023	gain	Start-year reference	0/5	11/59	-0.186
AlphaEarth V1 mosaic	2022–2023	gain	Concurrent reference	0/5	11/59	-0.186
AlphaEarth V1 mosaic	2022–2023	gain	Control-mean adjustment	0/5	13/59	-0.220

Representation	Pair end years	Task	Score / classifier	True positives / positives	False positives / negatives	J
AlphaEarth V1 mosaic	2022–2023	gain	Cosine change magnitude	0/5	0/59	0.000
AlphaEarth V1 mosaic	2022–2023	loss	Start-year reference	12/22	4/83	0.497
AlphaEarth V1 mosaic	2022–2023	loss	Concurrent reference	11/22	2/83	0.476
AlphaEarth V1 mosaic	2022–2023	loss	Control-mean adjustment	13/22	8/83	0.495
AlphaEarth V1 mosaic	2022–2023	loss	Cosine change magnitude	20/22	17/83	0.704
TESSERA v1	2022–2023	gain	Start-year reference	1/5	16/59	-0.071
TESSERA v1	2022–2023	gain	Concurrent reference	1/5	19/59	-0.122
TESSERA v1	2022–2023	gain	Control-mean adjustment	0/5	13/59	-0.220
TESSERA v1	2022–2023	gain	Cosine change magnitude	0/5	1/59	-0.017
TESSERA v1	2022–2023	loss	Start-year reference	21/22	20/83	0.714
TESSERA v1	2022–2023	loss	Concurrent reference	18/22	19/83	0.589
TESSERA v1	2022–2023	loss	Control-mean adjustment	18/22	9/83	0.710
TESSERA v1	2022–2023	loss	Cosine change magnitude	21/22	26/83	0.641
AlphaEarth V1 mosaic	2020–2021	gain	Start-year reference	2/13	5/60	0.071
AlphaEarth V1 mosaic	2020–2021	gain	Concurrent reference	0/13	4/60	-0.067
AlphaEarth V1 mosaic	2020–2021	gain	Control-mean adjustment	2/13	9/60	0.004
AlphaEarth V1 mosaic	2020–2021	gain	Cosine change magnitude	0/13	1/60	-0.017
AlphaEarth V1 mosaic	2020–2021	loss	Start-year reference	8/23	6/85	0.277
AlphaEarth V1 mosaic	2020–2021	loss	Concurrent reference	7/23	7/85	0.222
AlphaEarth V1 mosaic	2020–2021	loss	Control-mean adjustment	8/23	7/85	0.265
AlphaEarth V1 mosaic	2020–2021	loss	Cosine change magnitude	20/23	5/85	0.811
TESSERA v1	2020–2021	gain	Start-year reference	2/13	14/60	-0.079
TESSERA v1	2020–2021	gain	Concurrent reference	1/13	12/60	-0.123
TESSERA v1	2020–2021	gain	Control-mean adjustment	0/13	5/60	-0.083
TESSERA v1	2020–2021	gain	Cosine change magnitude	0/13	4/60	-0.067
TESSERA v1	2020–2021	loss	Start-year reference	12/23	14/85	0.357
TESSERA v1	2020–2021	loss	Concurrent reference	12/23	18/85	0.310
TESSERA v1	2020–2021	loss	Control-mean adjustment	11/23	14/85	0.314
TESSERA v1	2020–2021	loss	Cosine change magnitude	18/23	16/85	0.594

Appendix C. Saved single-date native input sensitivity

Only one date per pair is substituted; the other remains a mosaic vector. The fixed-pair native experiment remains incomplete. Identical-score tolerance is an absolute tolerance of 1e-12. Counts below refer to the same small set of pairs across multiple methods and are not independent observations. Full scores, common-case TP/FP counts and input identities are supplied in analysis/native-single-date-v22.

Support	Task	Score / classifier	Cases	Identical substituted inputs	Identical scores	Changed flags
centre	gain	Concurrent reference	2	2	2	0
centre	gain	Control-mean adjustment	2	2	2	0
centre	gain	Cosine change magnitude	2	2	2	0
centre	gain	Direct gradient boosting	2	2	2	0
centre	gain	End-year reference	2	2	2	0
centre	gain	Pooled reference	2	2	2	0
centre	gain	Standardized concurrent	2	2	2	0
centre	gain	Start-year reference	2	2	2	0
centre	gain	State random forest	2	2	2	0
centre	gain	Subsequent-reference similarity	2	2	2	0
centre	gain	Subsequent-state logistic	2	2	2	0
centre	gain	Subsequent-state forest	2	2	2	0
centre	loss	Concurrent reference	18	8	8	2
centre	loss	Control-mean adjustment	18	8	8	1
centre	loss	Cosine change magnitude	18	8	8	0
centre	loss	Direct gradient boosting	18	8	8	1
centre	loss	End-year reference	18	8	8	2
centre	loss	Pooled reference	18	8	8	2
centre	loss	Standardized concurrent	18	8	8	1
centre	loss	Start-year reference	18	8	8	2
centre	loss	State random forest	18	8	8	3
centre	loss	Subsequent-reference similarity	18	8	11	1
centre	loss	Subsequent-state logistic	18	8	11	0
centre	loss	Subsequent-state forest	18	8	11	0
patch	gain	Concurrent reference	2	1	1	0
patch	gain	Control-mean adjustment	2	1	1	0

Support	Task	Score / classifier	Cases	Identical substituted inputs	Identical scores	Changed flags
patch	gain	Cosine change magnitude	2	1	1	0
patch	gain	Direct gradient boosting	2	1	1	0
patch	gain	End-year reference	2	1	1	0
patch	gain	Pooled reference	2	1	1	0
patch	gain	Standardized concurrent	2	1	1	0
patch	gain	Start-year reference	2	1	1	0
patch	gain	State random forest	2	1	1	0
patch	gain	Subsequent-reference similarity	2	1	1	0
patch	gain	Subsequent-state logistic	2	1	1	0
patch	gain	Subsequent-state forest	2	1	1	0
patch	loss	Concurrent reference	18	6	6	1
patch	loss	Control-mean adjustment	18	6	6	1
patch	loss	Cosine change magnitude	18	6	6	0
patch	loss	Direct gradient boosting	18	6	9	0
patch	loss	End-year reference	18	6	6	1
patch	loss	Pooled reference	18	6	6	0
patch	loss	Standardized concurrent	18	6	6	2
patch	loss	Start-year reference	18	6	6	2
patch	loss	State random forest	18	6	6	0
patch	loss	Subsequent-reference similarity	18	6	10	1
patch	loss	Subsequent-state logistic	18	6	10	0
patch	loss	Subsequent-state forest	18	6	10	0

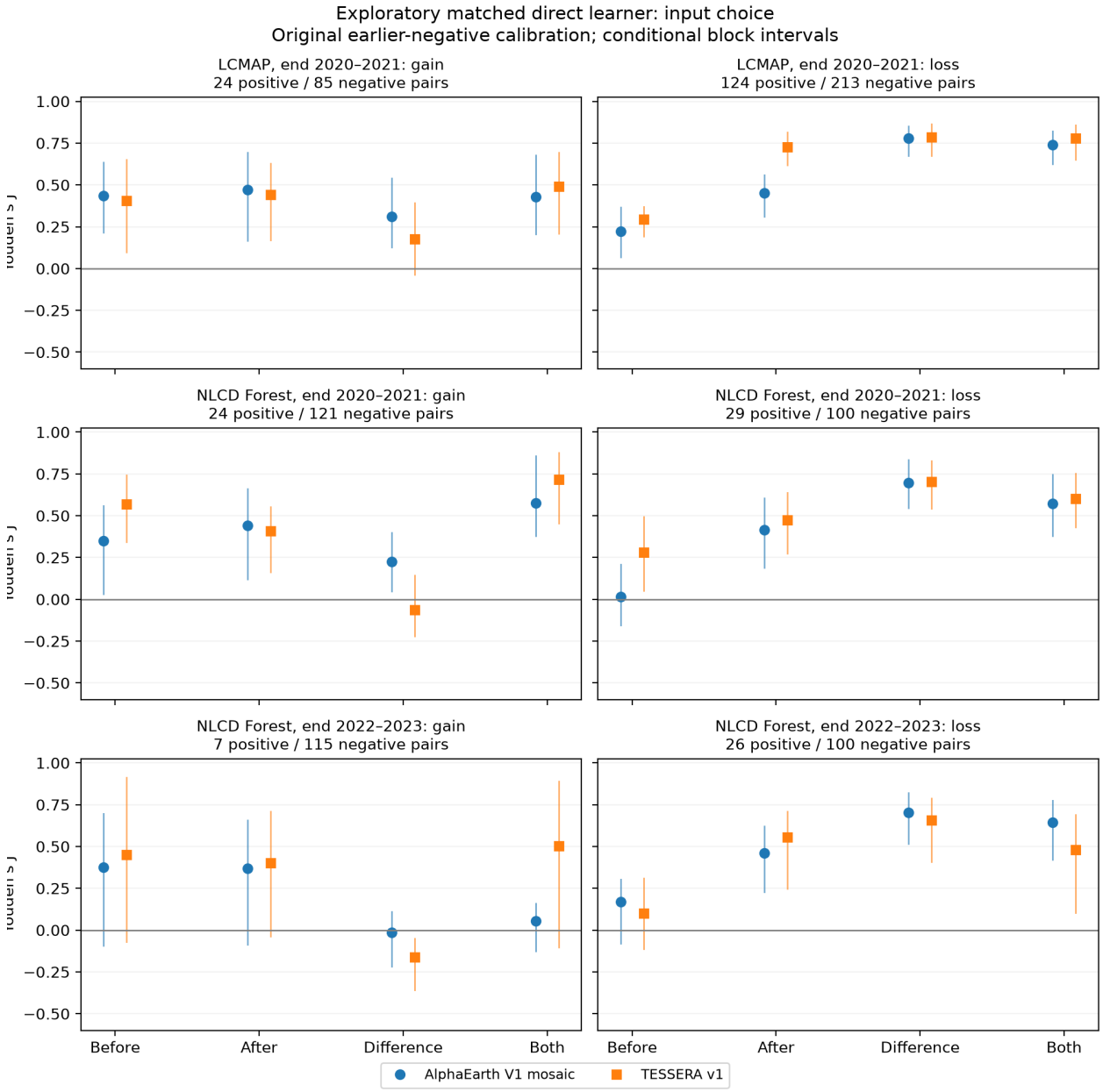
Appendix D. Matched direct-learner input comparisons

Retained paired contrasts: Matched direct-learner input comparison: exploratory

The initial-vector state diagnostic does not hold the training target and algorithm constant against the direct transition comparator. The later exploratory matched-learner analysis addresses that comparison: before-only, after-only, difference-only and original both-date inputs use the same earlier transition labels, learner hyperparameters, partitions and calibration rule. Input dimensions differ, so equal settings do not imply equal capacity. Original both-date scores/flags/thresholds are reconstructed exactly. Both endpoints had already been inspected before this amendment; the Forest results here are reused exploratory data, not a fresh validation.

The before-only input retains gain discrimination in the observed sample. Difference-only J is lower than both-date J in 6/6 gain conditions and higher in 6/6 loss conditions across the three separately reported populations and

two products. These are descriptive point directions on shared fitted samples; conditional intervals often include zero, and no universal ordering or equivalence is established. A learned difference-vector score or classifier can flag some LCMAP gains even when scalar magnitude flags none. Conversely, differencing alone does not give uniformly strong gain transfer. These results qualify interpretations based solely on either a magnitude score or a subsequent-state classifier.



Appendix D retains every input condition's operating point. The original LCMAP reduced-input contrasts below show uncertainty and changed decisions explicitly; the separate Forest contrasts and label-agreement subsets are in the supplied CSV files. Before-only results remain retrospective associations with dated vector values, not prospective forecasts or identification of the source of the information.

Representation	Task	Input versus both-date direct learner	J difference	Conditional 95% interval	Changed flags / cases
AlphaEarth V1 mosaic	gain	Direct learner: before only	0.006	-0.240 to 0.217	14/109
AlphaEarth V1 mosaic	loss	Direct learner: before only	-0.519	-0.687 to -0.355	123/337
TESSERA v1	gain	Direct learner: before only	-0.082	-0.229 to 0.080	15/109
TESSERA v1	loss	Direct learner: before only	-0.486	-0.585 to -0.379	102/337
AlphaEarth V1 mosaic	gain	Direct learner: after only	0.043	-0.348 to 0.346	22/109
AlphaEarth V1 mosaic	loss	Direct learner: after only	-0.289	-0.384 to -0.193	71/337
TESSERA v1	gain	Direct learner: after only	-0.048	-0.155 to 0.057	9/109
TESSERA v1	loss	Direct learner: after only	-0.051	-0.124 to 0.024	44/337
AlphaEarth V1 mosaic	gain	Direct learner: difference only	-0.119	-0.341 to 0.115	17/109
AlphaEarth V1 mosaic	loss	Direct learner: difference only	0.040	-0.053 to 0.139	51/337
TESSERA v1	gain	Direct learner: difference only	-0.313	-0.609 to 0.043	41/109
TESSERA v1	loss	Direct learner: difference only	0.007	-0.080 to 0.106	45/337

All four fixed inputs are shown. Both-date input means the original concatenation of before, after and difference vectors. This amendment was fixed after inspecting both endpoints; all rows are exploratory and share earlier LCMAP fitting/calibration. Full conditional paired intervals, year/region strata, feature dimensions and threshold/reconstruction checks are in analysis/direct-inputs-v23 and analysis/nlcd-direct-inputs-v23. No winning input condition is selected.

Test population	Representation	Task	Direct input condition	True positives / positives	False positives / negatives	J	AUROC
LCMAP, end 2020–2021	AlphaEarth V1 mosaic	gain	Direct learner: after only	15/24	13/85	0.472	0.861
LCMAP, end 2020–2021	AlphaEarth V1 mosaic	gain	Direct learner: before only	13/24	9/85	0.436	0.812
LCMAP, end 2020–2021	AlphaEarth V1 mosaic	gain	Direct learner: difference only	10/24	9/85	0.311	0.729
LCMAP, end 2020–2021	AlphaEarth V1 mosaic	gain	Direct gradient boosting	12/24	6/85	0.429	0.866
LCMAP, end 2020–2021	AlphaEarth V1 mosaic	loss	Direct learner: after only	74/124	31/213	0.451	0.847
LCMAP, end 2020–2021	AlphaEarth V1 mosaic	loss	Direct learner: before only	42/124	25/213	0.221	0.702
LCMAP, end 2020–2021	AlphaEarth V1 mosaic	loss	Direct learner: difference only	113/124	28/213	0.780	0.953
LCMAP, end 2020–2021	AlphaEarth V1 mosaic	loss	Direct gradient boosting	111/124	33/213	0.740	0.957
LCMAP, end 2020–2021	TESSERA v1	gain	Direct learner: after only	12/24	5/85	0.441	0.876
LCMAP, end 2020–2021	TESSERA v1	gain	Direct learner: before only	14/24	15/85	0.407	0.853
LCMAP, end 2020–2021	TESSERA v1	gain	Direct learner: difference only	11/24	24/85	0.176	0.621
LCMAP, end 2020–2021	TESSERA v1	gain	Direct gradient boosting	14/24	8/85	0.489	0.874

Test population	Representation	Task	Direct input condition	True positives / positives	False positives / negatives	J	AUROC
LCMAP, end 2020–2021	TESSERA v1	loss	Direct learner: after only	110/124	34/213	0.727	0.932
LCMAP, end 2020–2021	TESSERA v1	loss	Direct learner: before only	59/124	39/213	0.293	0.764
LCMAP, end 2020–2021	TESSERA v1	loss	Direct learner: difference only	109/124	20/213	0.785	0.948
LCMAP, end 2020–2021	TESSERA v1	loss	Direct gradient boosting	114/124	30/213	0.779	0.959
NLCD Forest, end 2022–2023	AlphaEarth V1 mosaic	gain	Direct learner: after only	3/7	7/115	0.368	0.834
NLCD Forest, end 2022–2023	AlphaEarth V1 mosaic	gain	Direct learner: before only	3/7	6/115	0.376	0.831
NLCD Forest, end 2022–2023	AlphaEarth V1 mosaic	gain	Direct learner: difference only	1/7	18/115	-0.014	0.544
NLCD Forest, end 2022–2023	AlphaEarth V1 mosaic	gain	Direct gradient boosting	1/7	10/115	0.056	0.793
NLCD Forest, end 2022–2023	AlphaEarth V1 mosaic	loss	Direct learner: after only	19/26	27/100	0.461	0.844
NLCD Forest, end 2022–2023	AlphaEarth V1 mosaic	loss	Direct learner: before only	8/26	14/100	0.168	0.670
NLCD Forest, end 2022–2023	AlphaEarth V1 mosaic	loss	Direct learner: difference only	25/26	26/100	0.702	0.940
NLCD Forest, end 2022–2023	AlphaEarth V1 mosaic	loss	Direct gradient boosting	24/26	28/100	0.643	0.920
NLCD Forest, end 2022–2023	TESSERA v1	gain	Direct learner: after only	3/7	3/115	0.402	0.771
NLCD Forest, end 2022–2023	TESSERA v1	gain	Direct learner: before only	4/7	14/115	0.450	0.831
NLCD Forest, end 2022–2023	TESSERA v1	gain	Direct learner: difference only	1/7	35/115	-0.161	0.328
NLCD Forest, end 2022–2023	TESSERA v1	gain	Direct gradient boosting	4/7	8/115	0.502	0.860
NLCD Forest, end 2022–2023	TESSERA v1	loss	Direct learner: after only	22/26	29/100	0.556	0.867
NLCD Forest, end 2022–2023	TESSERA v1	loss	Direct learner: before only	7/26	17/100	0.099	0.535
NLCD Forest, end 2022–2023	TESSERA v1	loss	Direct learner: difference only	22/26	19/100	0.656	0.882
NLCD Forest, end 2022–2023	TESSERA v1	loss	Direct gradient boosting	19/26	25/100	0.481	0.868
NLCD Forest, end 2020–2021	AlphaEarth V1 mosaic	gain	Direct learner: after only	13/24	12/121	0.442	0.890
NLCD Forest, end 2020–2021	AlphaEarth V1 mosaic	gain	Direct learner: before only	10/24	8/121	0.351	0.857
NLCD Forest, end 2020–2021	AlphaEarth V1 mosaic	gain	Direct learner: difference only	8/24	13/121	0.226	0.722
NLCD Forest, end 2020–2021	AlphaEarth V1 mosaic	gain	Direct gradient boosting	15/24	6/121	0.575	0.917
NLCD Forest, end 2020–2021	AlphaEarth V1 mosaic	loss	Direct learner: after only	19/29	24/100	0.415	0.790
NLCD Forest, end 2020–2021	AlphaEarth V1 mosaic	loss	Direct learner: before only	6/29	19/100	0.017	0.609
NLCD Forest, end 2020–2021	AlphaEarth V1 mosaic	loss	Direct learner: difference only	24/29	13/100	0.698	0.903
NLCD Forest, end 2020–2021	AlphaEarth V1 mosaic	loss	Direct gradient boosting	23/29	22/100	0.573	0.884
NLCD Forest, end 2020–2021	TESSERA v1	gain	Direct learner: after only	11/24	6/121	0.409	0.914
NLCD Forest, end 2020–2021	TESSERA v1	gain	Direct learner: before only	17/24	17/121	0.568	0.903
NLCD Forest, end 2020–2021	TESSERA v1	gain	Direct learner: difference only	8/24	48/121	-0.063	0.446

Test population	Representation	Task	Direct input condition	True positives / positives	False positives / negatives	J	AUROC
NLCD Forest, end 2020–2021	TESSERA v1	gain	Direct gradient boosting	19/24	9/121	0.717	0.936
NLCD Forest, end 2020–2021	TESSERA v1	loss	Direct learner: after only	23/29	32/100	0.473	0.804
NLCD Forest, end 2020–2021	TESSERA v1	loss	Direct learner: before only	16/29	27/100	0.282	0.621
NLCD Forest, end 2020–2021	TESSERA v1	loss	Direct learner: difference only	23/29	9/100	0.703	0.913
NLCD Forest, end 2020–2021	TESSERA v1	loss	Direct gradient boosting	23/29	19/100	0.603	0.861

Appendix E. Reference/control availability and longer reference intervals

Post-review descriptive diagnostics. All counts refer to baseline-selected plots outside the named fold, not verified later stability. Full one-year/individual-plot rows and same-membership means are retained in analysis/review-diagnostics-r1. No multi-year target screening is inferred.

Representation	Held-out region	Year	Assigned references outside region	Valid references	Assigned controls outside region	Valid controls
AlphaEarth V1 mosaic	0	2017	53	53	51	51
AlphaEarth V1 mosaic	0	2018	53	53	51	51
AlphaEarth V1 mosaic	0	2019	53	53	51	51
AlphaEarth V1 mosaic	0	2020	53	53	51	51
AlphaEarth V1 mosaic	0	2021	53	53	51	51
AlphaEarth V1 mosaic	0	2022	53	53	51	51
AlphaEarth V1 mosaic	0	2023	53	53	51	51
AlphaEarth V1 mosaic	1	2017	49	49	46	46
AlphaEarth V1 mosaic	1	2018	49	49	46	46
AlphaEarth V1 mosaic	1	2019	49	49	46	46
AlphaEarth V1 mosaic	1	2020	49	49	46	46
AlphaEarth V1 mosaic	1	2021	49	49	46	46
AlphaEarth V1 mosaic	1	2022	49	49	46	46
AlphaEarth V1 mosaic	1	2023	49	49	46	46
AlphaEarth V1 mosaic	2	2017	48	48	53	53
AlphaEarth V1 mosaic	2	2018	48	48	53	53
AlphaEarth V1 mosaic	2	2019	48	48	53	53
AlphaEarth V1 mosaic	2	2020	48	48	53	53
AlphaEarth V1 mosaic	2	2021	48	48	53	53

Representation	Held-out region	Year	Assigned references outside region	Valid references	Assigned controls outside region	Valid controls
AlphaEarth V1 mosaic	2	2022	48	48	53	53
AlphaEarth V1 mosaic	2	2023	48	48	53	53
AlphaEarth V1 mosaic	3	2017	39	39	38	38
AlphaEarth V1 mosaic	3	2018	39	39	38	38
AlphaEarth V1 mosaic	3	2019	39	39	38	38
AlphaEarth V1 mosaic	3	2020	39	39	38	38
AlphaEarth V1 mosaic	3	2021	39	39	38	38
AlphaEarth V1 mosaic	3	2022	39	39	38	38
AlphaEarth V1 mosaic	3	2023	39	39	38	38
AlphaEarth V1 mosaic	4	2017	43	43	44	44
AlphaEarth V1 mosaic	4	2018	43	43	44	44
AlphaEarth V1 mosaic	4	2019	43	43	44	44
AlphaEarth V1 mosaic	4	2020	43	43	44	44
AlphaEarth V1 mosaic	4	2021	43	43	44	44
AlphaEarth V1 mosaic	4	2022	43	43	44	44
AlphaEarth V1 mosaic	4	2023	43	43	44	44
TESSERA v1	0	2017	53	53	51	50
TESSERA v1	0	2018	53	51	51	47
TESSERA v1	0	2019	53	53	51	50
TESSERA v1	0	2020	53	53	51	50
TESSERA v1	0	2021	53	53	51	50
TESSERA v1	0	2022	53	53	51	50
TESSERA v1	0	2023	53	53	51	50
TESSERA v1	1	2017	49	49	46	45
TESSERA v1	1	2018	49	47	46	42
TESSERA v1	1	2019	49	49	46	45
TESSERA v1	1	2020	49	49	46	45
TESSERA v1	1	2021	49	49	46	45
TESSERA v1	1	2022	49	49	46	45
TESSERA v1	1	2023	49	49	46	45
TESSERA v1	2	2017	48	48	53	52
TESSERA v1	2	2018	48	46	53	49
TESSERA v1	2	2019	48	48	53	52
TESSERA v1	2	2020	48	48	53	52
TESSERA v1	2	2021	48	48	53	52
TESSERA v1	2	2022	48	48	53	52
TESSERA v1	2	2023	48	48	53	52
TESSERA v1	3	2017	39	39	38	37
TESSERA v1	3	2018	39	37	38	34
TESSERA v1	3	2019	39	39	38	37
TESSERA v1	3	2020	39	39	38	37
TESSERA v1	3	2021	39	39	38	37
TESSERA v1	3	2022	39	39	38	37
TESSERA v1	3	2023	39	39	38	37

Representation	Held-out region	Year	Assigned references outside region	Valid references	Assigned controls outside region	Valid controls
TESSERA v1	4	2017	43	43	44	44
TESSERA v1	4	2018	43	43	44	44
TESSERA v1	4	2019	43	43	44	44
TESSERA v1	4	2020	43	43	44	44
TESSERA v1	4	2021	43	43	44	44
TESSERA v1	4	2022	43	43	44	44
TESSERA v1	4	2023	43	43	44	44

Descriptive 2017-to-later reference intervals:

Representation	Held-out region	Reference interval	References before	References after	Paired controls	Centroid dissimilarity	Median control dissimilarity
AlphaEarth V1 mosaic	0	2017–2019	53	53	51	0.0098	0.0297
AlphaEarth V1 mosaic	0	2017–2020	53	53	51	0.0065	0.0239
AlphaEarth V1 mosaic	0	2017–2021	53	53	51	0.0068	0.0297
AlphaEarth V1 mosaic	0	2017–2022	53	53	51	0.0112	0.0366
AlphaEarth V1 mosaic	0	2017–2023	53	53	51	0.0101	0.0339
AlphaEarth V1 mosaic	1	2017–2019	49	49	46	0.0104	0.0258
AlphaEarth V1 mosaic	1	2017–2020	49	49	46	0.0067	0.0263
AlphaEarth V1 mosaic	1	2017–2021	49	49	46	0.0078	0.0275
AlphaEarth V1 mosaic	1	2017–2022	49	49	46	0.0133	0.0316
AlphaEarth V1 mosaic	1	2017–2023	49	49	46	0.0105	0.0330
AlphaEarth V1 mosaic	2	2017–2019	48	48	53	0.0089	0.0253
AlphaEarth V1 mosaic	2	2017–2020	48	48	53	0.0056	0.0255
AlphaEarth V1 mosaic	2	2017–2021	48	48	53	0.0069	0.0282
AlphaEarth V1 mosaic	2	2017–2022	48	48	53	0.0116	0.0334
AlphaEarth V1 mosaic	2	2017–2023	48	48	53	0.0095	0.0333
AlphaEarth V1 mosaic	3	2017–2019	39	39	38	0.0118	0.0247
AlphaEarth V1 mosaic	3	2017–2020	39	39	38	0.0056	0.0266
AlphaEarth V1 mosaic	3	2017–2021	39	39	38	0.0087	0.0283
AlphaEarth V1 mosaic	3	2017–2022	39	39	38	0.0127	0.0366
AlphaEarth V1 mosaic	3	2017–2023	39	39	38	0.0092	0.0333
AlphaEarth V1 mosaic	4	2017–2019	43	43	44	0.0160	0.0325
AlphaEarth V1 mosaic	4	2017–2020	43	43	44	0.0086	0.0280
AlphaEarth V1 mosaic	4	2017–2021	43	43	44	0.0079	0.0355

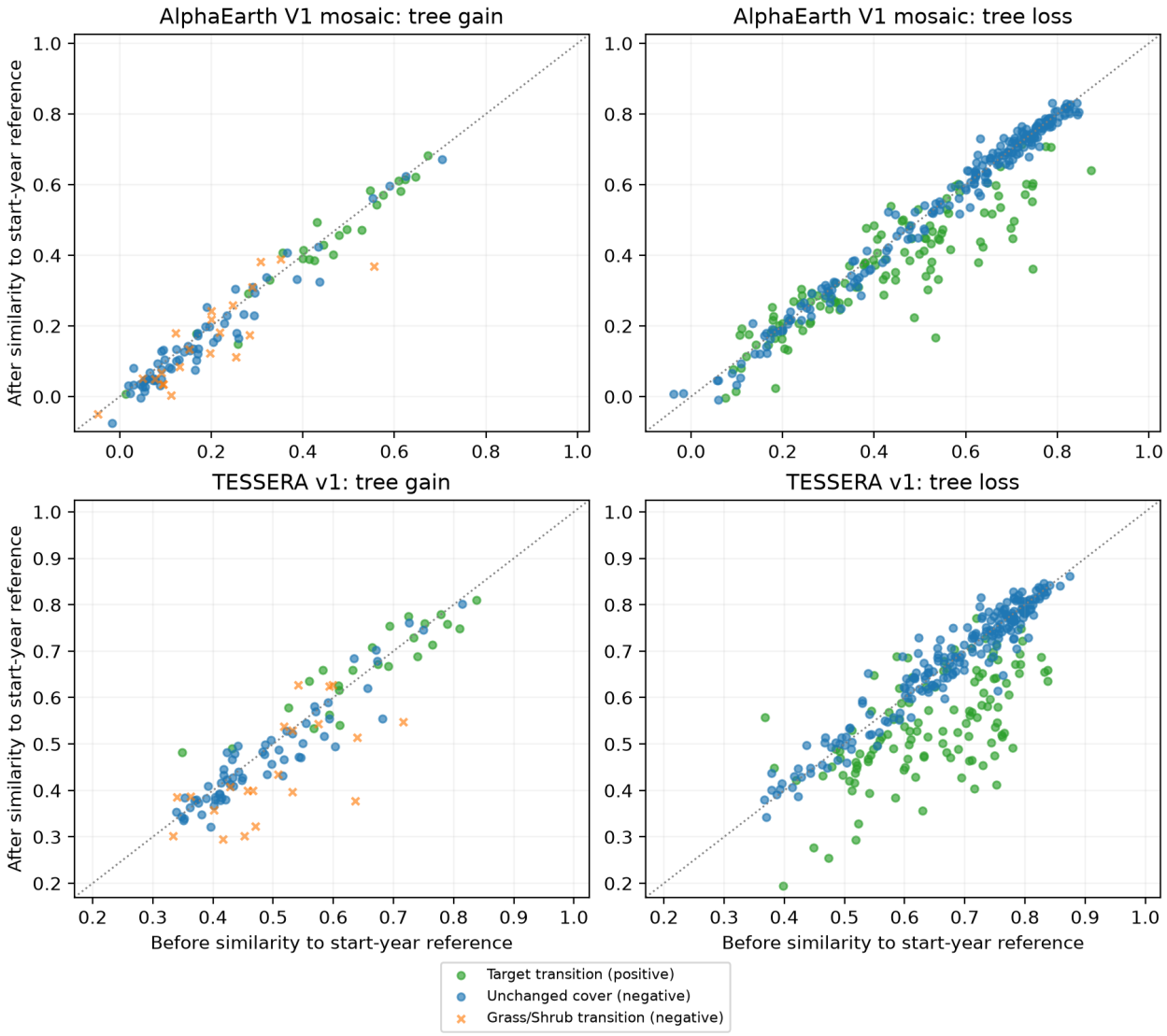
Representation	Held-out region	Reference interval	References before	References after	Paired controls	Centroid dissimilarity	Median control dissimilarity
AlphaEarth V1 mosaic	4	2017–2022	43	43	44	0.0177	0.0371
AlphaEarth V1 mosaic	4	2017–2023	43	43	44	0.0153	0.0366
TESSERA v1	0	2017–2019	53	53	50	0.0541	0.1426
TESSERA v1	0	2017–2020	53	53	50	0.0601	0.1537
TESSERA v1	0	2017–2021	53	53	50	0.0720	0.1547
TESSERA v1	0	2017–2022	53	53	50	0.0835	0.1735
TESSERA v1	0	2017–2023	53	53	50	0.0758	0.1739
TESSERA v1	1	2017–2019	49	49	45	0.0566	0.1429
TESSERA v1	1	2017–2020	49	49	45	0.0626	0.1625
TESSERA v1	1	2017–2021	49	49	45	0.0748	0.1627
TESSERA v1	1	2017–2022	49	49	45	0.0845	0.1773
TESSERA v1	1	2017–2023	49	49	45	0.0790	0.1741
TESSERA v1	2	2017–2019	48	48	52	0.0543	0.1390
TESSERA v1	2	2017–2020	48	48	52	0.0609	0.1481
TESSERA v1	2	2017–2021	48	48	52	0.0697	0.1547
TESSERA v1	2	2017–2022	48	48	52	0.0840	0.1735
TESSERA v1	2	2017–2023	48	48	52	0.0722	0.1706
TESSERA v1	3	2017–2019	39	39	37	0.0523	0.1516
TESSERA v1	3	2017–2020	39	39	37	0.0625	0.1568
TESSERA v1	3	2017–2021	39	39	37	0.0774	0.1833
TESSERA v1	3	2017–2022	39	39	37	0.0808	0.1903
TESSERA v1	3	2017–2023	39	39	37	0.0790	0.1889
TESSERA v1	4	2017–2019	43	43	44	0.0530	0.1300
TESSERA v1	4	2017–2020	43	43	44	0.0560	0.1403
TESSERA v1	4	2017–2021	43	43	44	0.0654	0.1650
TESSERA v1	4	2017–2022	43	43	44	0.0831	0.1664
TESSERA v1	4	2017–2023	43	43	44	0.0721	0.1625

Appendix F. Retained exploratory and unfinished checks

Endpoint composition and reference availability

The gain-negative definition includes 21 later Grass/Shrub transitions. Removing them from this separate calibration/test condition leaves 64 unchanged-cover negatives and 24 gain positives. Cosine change magnitude still detects 0 gains for AlphaEarth; the full two-product table is analysis/endpoint-v13/stable-negative-gain-table.md. Fitted scores are unchanged on all common cases. Within this sample, the floor persists after removing those transitions. Its biological, temporal or spatial-support cause remains unresolved.

Before/after similarity components reconstructed from saved scores agree with direct raw-vector calculations to a maximum absolute error of 5.55e-16. The scalar plot shows where the change score or classifier differs from an absolute subsequent-state score. It does not measure ecological condition or establish that either the labels or representations are wrong.



The same-reference-plots check changes the TESSERA gain concurrent-reference J contrast from -0.012 to +0.024. Target test scores remain fixed; earlier calibration inputs and thresholds change. Original separately-valid reference means remain primary. This sensitivity limits claims about a general advantage or disadvantage of concurrent references.

Additional controls: Alternative controls, calendar anchors and scores and classifiers

The preplanned non-tree, initial-class and frozen-calendar controls produce different operating points (Appendix A). Exact initial-class matching has fewer available cases, so its contrast must use the common-case denominators in the paired comparison file. A gain in its J cannot be compared by subtracting the full-sample fixed start-year reference J. No control rule gives a consistent advantage across all products, tasks and available cases.

The preplanned reference allocations, centre/patch supports and global/nearest-reference conditions give the ranges below. These settings reuse the same targets and pool; their signs are sensitivity counts, not independent evidence or a vote for a method. All settings remain available in the machine-readable tables.

Representation	Task	Score / classifier versus fixed start-year reference	Sensitivity settings	Minimum J difference	Maximum J difference	Positive	Negative	Zero
AlphaEarth V1 mosaic	gain	Concurrent reference	20	-0.095	0.119	9	10	1
AlphaEarth V1 mosaic	gain	Control-mean adjustment	20	-0.119	0.141	13	7	0
AlphaEarth V1 mosaic	loss	Concurrent reference	20	-0.169	0.003	1	19	0
AlphaEarth V1 mosaic	loss	Control-mean adjustment	20	-0.040	0.052	6	14	0
TESSERA v1	gain	Concurrent reference	20	-0.284	0.059	4	15	1
TESSERA v1	gain	Control-mean adjustment	20	-0.167	0.148	13	6	1
TESSERA v1	loss	Concurrent reference	20	-0.080	0.026	6	14	0
TESSERA v1	loss	Control-mean adjustment	20	-0.053	0.029	9	10	1

The original strict reference-region exclusion places the eighth-nearest reference a median 1,071 km away in the centre-pixel sample. An exploratory condition allowing baseline reference/control plots from all regions reduces that median to 579 km. Target fitting and calibration remain region-held-out; the reference-label information access changes. The concurrent-reference J contrast is negative in 8/8 centre conditions across global/nearest selection, while control effects vary (Appendix A). Supervised scores and flags are unchanged by construction and verified. References remain geographically broad and are not locally verified reference ecosystems, even under the relaxed access condition.

Exploratory baseline-cohort state classifiers use only the baseline reference/tree-control/non-tree-control cohorts for state fitting, while their thresholds still require earlier target-negative labels. They therefore reduce one source of state-fitting labels without establishing equal annotation budgets or zero transition-label dependence. Embedding-matched controls use initial-vector similarity rather than geographic distance. Their complete results are in Appendix A. Calibration at the additional five- and twenty-percent nominal levels changes some primary contrast directions; the original ten-percent rule remains primary.

The auxiliary classification release supplies 109 later labels among 166 baseline-cohort plots, with 0 known binary disagreements and 57 unknown later labels. These are isolated annual observations, not complete histories. No primary plot is removed using those later labels.

Input-period, coordinate and class checks: What requires the subsequent target vector? Exploratory diagnostics

Some gain discrimination remains when a state classifier receives only the initial target vector. The table compares the same fitted state model applied to the before and after vectors, each calibrated using its own earlier-negative scores. Both comparisons retain the same availability mask, which requires both products in both years. The state models were fitted using earlier labelled states, and the encoders and labels were produced retrospectively. These results are therefore associations in a dated representation, not prospective forecasts.

Representation	Task	State fit	Before vector TP; FP	After vector TP; FP	Before J	After J	Before minus after J	Conditional 95% interval	Changed flags / cases
AlphaEarth V1 mosaic	gain	Initial-vector state forest	11/24; 9/85	9/24; 9/85	0.352	0.269	0.083	-0.093 to 0.221	8/109
AlphaEarth V1 mosaic	loss	Initial-vector state forest	47/124; 56/213	113/124; 49/213	0.116	0.681	-0.565	-0.693 to -0.386	77/337
TESSERA v1	gain	Initial-vector state forest	14/24; 10/85	14/24; 5/85	0.466	0.525	-0.059	-0.200 to 0.070	11/109
TESSERA v1	loss	Initial-vector state forest	38/124; 49/213	119/124; 45/213	0.076	0.748	-0.672	-0.795 to -0.492	93/337
AlphaEarth V1 mosaic	gain	Initial-vector cohort forest	7/24; 6/85	7/24; 8/85	0.221	0.198	0.024	-0.092 to 0.148	4/109
AlphaEarth V1 mosaic	loss	Initial-vector cohort forest	44/124; 52/213	83/124; 38/213	0.111	0.491	-0.380	-0.471 to -0.268	63/337
TESSERA v1	gain	Initial-vector cohort forest	11/24; 9/85	15/24; 4/85	0.352	0.578	-0.225	-0.441 to 0.066	15/109
TESSERA v1	loss	Initial-vector cohort forest	26/124; 26/213	90/124; 34/213	0.088	0.566	-0.479	-0.569 to -0.374	78/337

Changing the subsequent target-vector values in one fixed within-year permutation leaves all 3568 initial-only scored rows unchanged across 446 target keys. This verifies input dependence in the implementation; it is not another independent empirical replication. Initial reference similarity and baseline-cohort logistic counterparts, including their uncertainty, are retained in analysis/initial-vector-v17/initial-versus-subsequent.csv.

The after vector improves several gain scores and classifiers and improves the full state forest substantially for losses. Thus these results do not support the claim that all apparent change accuracy comes from initial information. They do show why subsequent-state classification accuracy alone cannot establish how much information about a transition was newly observed after the initial date.

The separate coordinate-only control uses four sine/cosine latitude/longitude features, with no embedding values, year or initial vegetation subclass as learner inputs. It yields the following associations on the same conditional sample. This is one geographic baseline, not two product replications. It cannot identify the source of information in the embeddings; weaker results would not exclude richer static landscape information.

Task	Coordinate-only score or classifier	True positives / positives	False positives / negatives	J	AUROC
gain	Coordinate transition boosting	7/24	5/85	0.233	0.746
gain	Coordinate state forest	1/24	1/85	0.030	0.680
gain	Coordinate state logistic	4/24	1/85	0.155	0.722
loss	Coordinate transition boosting	91/124	88/213	0.321	0.656
loss	Coordinate state forest	45/124	50/213	0.128	0.568
loss	Coordinate state logistic	40/124	46/213	0.107	0.693

The exploratory initial-class-rate baseline flags 0/24 gains and 0/85 negatives; 62 gain-task cases tie their regional threshold. This simple baseline does not explain the initial-vector forest results: gains are flagged within both initial vegetation classes, as shown below. The original mixed-class thresholds are retained within each subgroup. Small subgroup denominators and richer interactions with landscape context remain limitations.

Representation	Initial vegetation class	State score or classifier	True positives / positives	False positives / negatives	J	AUROC
AlphaEarth V1 mosaic	Grass/forb/herb	Initial-vector state forest	3/9	4/40	0.233	0.794
AlphaEarth V1 mosaic	Grass/forb/herb	Subsequent-state forest	3/9	5/40	0.208	0.861
TESSERA v1	Grass/forb/herb	Initial-vector state forest	6/9	5/40	0.542	0.883
TESSERA v1	Grass/forb/herb	Subsequent-state forest	6/9	4/40	0.567	0.931
AlphaEarth V1 mosaic	Shrubs	Initial-vector state forest	8/15	5/45	0.422	0.769
AlphaEarth V1 mosaic	Shrubs	Subsequent-state forest	6/15	4/45	0.311	0.837
TESSERA v1	Shrubs	Initial-vector state forest	8/15	5/45	0.422	0.744
TESSERA v1	Shrubs	Subsequent-state forest	8/15	1/45	0.511	0.834

All 337 loss-task cases begin as Trees. The class-rate score is constant within each regional fit, making zero loss flags a consequence of the task and threshold definition. This is not counted as a discovery. Region-specific estimates can differ, so the pooled score is not globally constant.

Native-grid diagnostics

Of 120 fixed sampled point-years, 113 native patches were obtained, 113 centres are valid in both inputs and 49 raw centre vectors are exactly equal. Overall summaries are below; role/year/latitude strata and every selected slot remain in analysis/native-v5. Pixel-centre distances are in metres. Centre/patch cosine values compare representations at the same requested coordinates but potentially different footprints.

Native/mosaic co-location metric	Valid pairs	Minimum	Fifth percentile	Median	Maximum
Centre cosine	113	0.907660	0.982823	0.999399	1.000000
Patch cosine	113	0.977003	0.991166	0.999435	1.000000
Pixel-centre distance (m)	113	0.304594	1.364329	5.021877	9.867387

This diagnostic neither re-fits the full study on native inputs nor tests stability of every small threshold contrast. Any disagreement confines the AlphaEarth estimates to the geographic mosaic actually evaluated. Cross-product differences cannot be attributed solely to encoders with their different input-processing paths.

The separately fixed paired-native case check is unfinished: only 1/32 point-years are available, leaving 0/16 complete pairs. Its empty counts are not evidence of unchanged decisions. Checking all saved original targets found no complete native pair, so the full paired comparison cannot be recovered from these measurements without additional acquisition.

An exploratory saved-data alternative replaces one available target date in each of 20 pairs, leaving its other date as a mosaic input. It contains 1 gain positive/1 gain negative and 10 loss positive/8 loss negatives; 6 substitutions affect the before date and 14 the after date. All fitting, references, controls and calibration remain mosaics. At least one flag changes in 5/20 pairs across the retained scores and classifiers and supports. This is an availability-selected, mixed-footprint case test, not an estimate of full native-grid performance or which footprint is correct.

Identity counts added during self-check show 10/20 identical centre vectors and 7/20 identical patch vectors. Both gain-case centre inputs are identical, so their unchanged centre decisions are predictable, not robustness evidence. Subsequent-state scores can also stay fixed when only the initial date changes. Appendix C gives these counts for every score or classifier and separates score identity from changed flags. The observed changes reinforce the geographic-mosaic scope of small AlphaEarth contrasts; paired-native transfer remains untested.

Partition sensitivity: Earlier fitting/calibration partition sensitivity: exploratory

The final exploratory check completed all 10 fixed earlier geographic-cell assignments: the original partition and its 9 perturbations. Reference cohorts, model settings, later tests and the calibration rule remain fixed. The original condition reconstructs 21408 prediction rows and 480 thresholds. Unsupervised score values on common keys are unchanged; their transferred decisions can vary through calibration. Learned scores can also vary through fitting.

The table retains every centre primary correction and direct-learner contrast. Its range is over these overlapping conditions, not an uncertainty interval or a selection of a best split. Positive/negative direction counts are not independent replications. Case ranges show whether differing availability could explain contrasts. There are 4800/4800 available method/fold conditions across all scores and classifiers/supports; 0 lack a threshold record and 0 have invalid calibration. Unavailable conditions are retained in the output rather than replaced.

Representation	Task	Score / classifier versus fixed start-year reference	Available partitions	J difference range	Positive/zero/negative directions	Positive / negative case ranges	Changed flags range
AlphaEarth V1 mosaic	gain	Concurrent reference	10	-0.035 to +0.065	5/3/2	24–24 / 85–85	0–6
AlphaEarth V1 mosaic	gain	Control-mean adjustment	10	-0.053 to +0.030	2/5/3	24–24 / 85–85	0–9
AlphaEarth V1 mosaic	gain	Direct gradient boosting	10	+0.310 to +0.643	10/0/0	24–24 / 85–85	14–41
AlphaEarth V1 mosaic	loss	Concurrent reference	10	-0.055 to +0.015	1/0/9	124–124 / 213–213	16–34
AlphaEarth V1 mosaic	loss	Control-mean adjustment	10	-0.069 to +0.026	1/0/9	124–124 / 213–213	6–24
AlphaEarth V1 mosaic	loss	Direct gradient boosting	10	+0.191 to +0.352	10/0/0	124–124 / 213–213	69–134
TESSERA v1	gain	Concurrent reference	10	-0.071 to +0.072	2/1/7	24–24 / 85–85	0–10
TESSERA v1	gain	Control-mean adjustment	10	-0.072 to +0.065	6/1/3	24–24 / 85–85	1–10
TESSERA v1	gain	Direct gradient boosting	10	+0.256 to +0.484	10/0/0	24–24 / 85–85	26–52
TESSERA v1	loss	Concurrent reference	10	-0.065 to +0.059	3/0/7	124–124 / 213–213	10–38
TESSERA v1	loss	Control-mean adjustment	10	-0.018 to +0.064	4/0/6	124–124 / 213–213	3–23
TESSERA v1	loss	Direct gradient boosting	10	-0.124 to +0.208	6/0/4	124–124 / 213–213	64–92

Magnitude-score gain true-positive ranges across these same partitions are AlphaEarth V1 mosaic centre: 0–1; AlphaEarth V1 mosaic patch: 0–0; TESSERA v1 centre: 0–0; TESSERA v1 patch: 0–0. The nonzero AlphaEarth centre condition is seed 14015: 1/24 true positives and 17/85 false positives. This bounds the original zero-count observation; the original partition is not replaced.

All partition-specific counts, operating points, conditional intervals, cell memberships, fitting counts and patch-support counterparts remain in analysis/partition-sensitivity-v24. This check examines one fitting/calibration choice; it does not estimate independent training-data, encoder or future-year variance. The original results remain primary.

Revision history

Revised in response to review-014-1: narrow the title/question to CONUS one-year tests; shorten the abstract; add primary AUROC, exact-fold reference/cohort variation, ranking similarity and year/region denominators; foreground the gain-ranking inversion; qualify AlphaEarth training-target and production-version dependence; move secondary/exploratory checks to appendices. All original scores, thresholds, failures and first-draft snapshots are retained. New saved-data diagnostics are exploratory and fixed after review in protocol/review-analysis-v1.md. Point-by-point decisions, access limits and change records are in review/response-1.md.